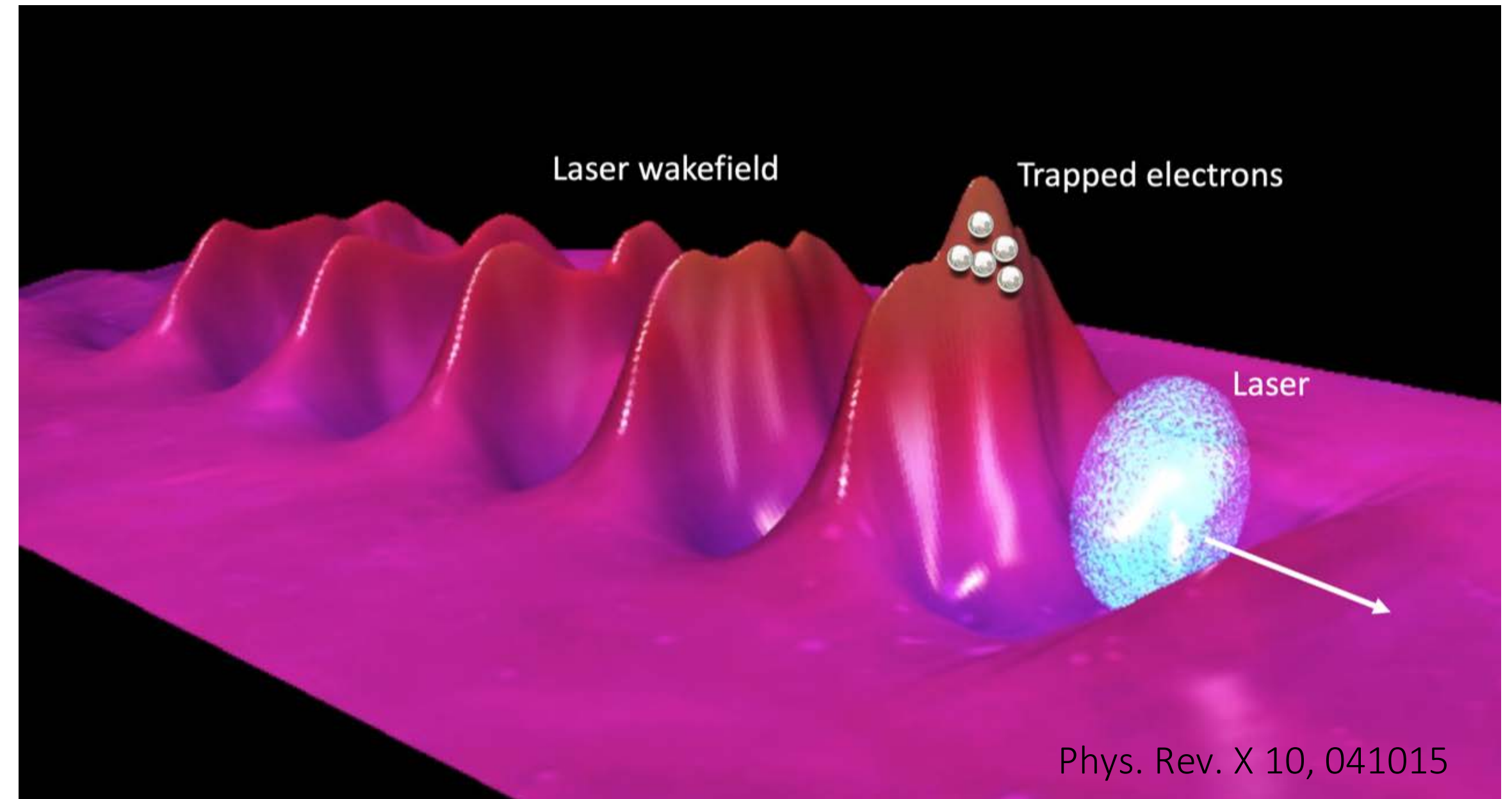
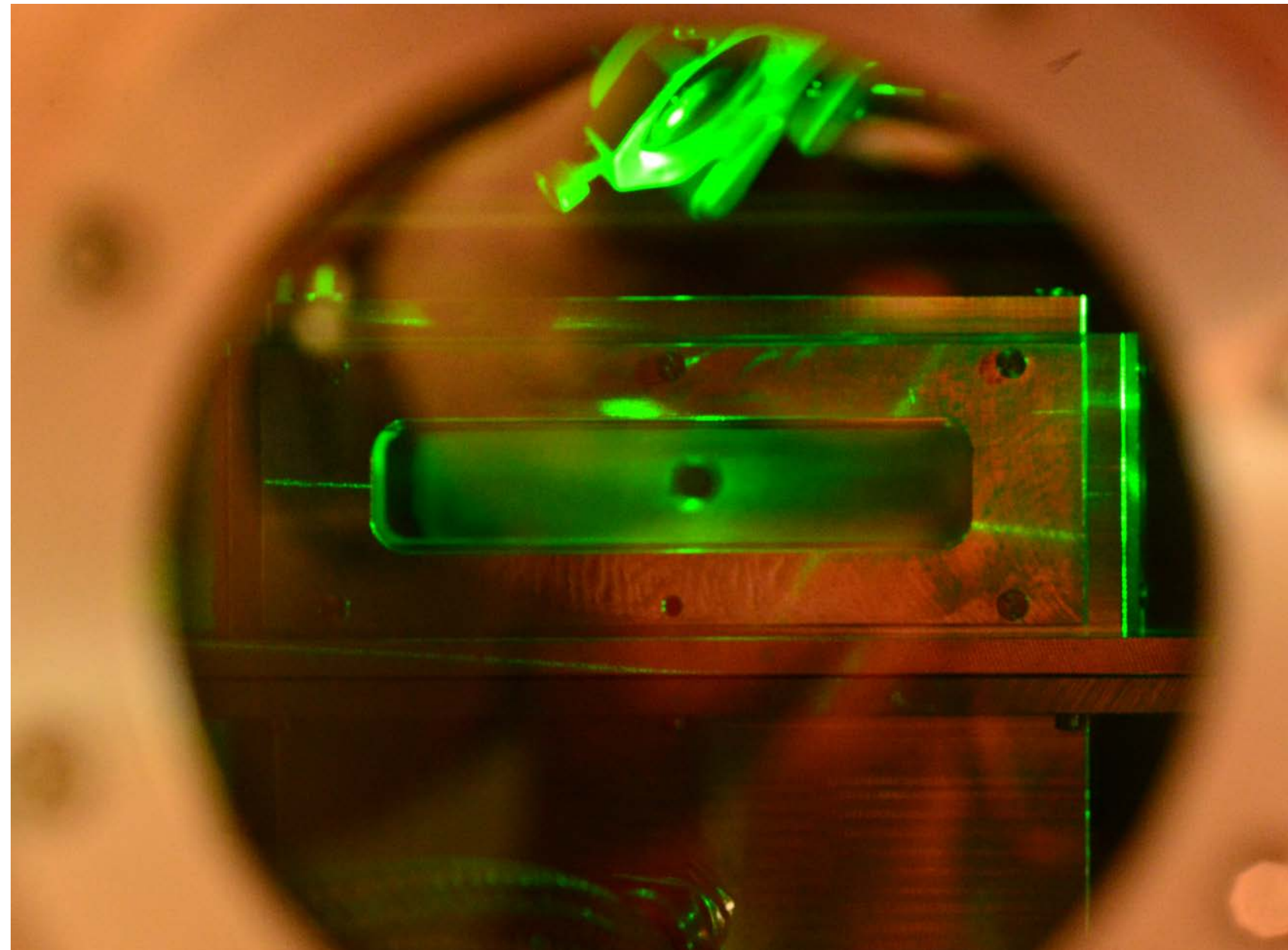


High charge, >10 GeV laser plasma electron acceleration

-

Design considerations for laser-accelerators and -XFELs



*B. M. Hegelich
University of Texas at Austin*

-

NAPAC 2022, Albuquerque, August 10, 2022

Acknowledgements

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- B. M. Hegelich, M. Downer



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- Constantin Aniculaesei, Ou Zhang Labun, Lance Labun, Rafal Zsagad

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- Ritwik, Sain, Eddie McCary, Drue Hood-McFadden, Austin Gottfredson, Xuejing Jiao, Ganesh Tiwari, Scott Luedtke, Paul King, Andrea Hannash



- GIST/CoReLS Gwangju:

Constantin Aniculaesei, Cheonha Jeon, Hyung-Taek Kim, Il Woo Choi, Yong Joo Rhee, Bo Ram Lee, Prashant Singh, Seung Woo Kang, Seung Jeon Kim, Tae Yoon Kim, Sooyang Lee, Chang Hee Nam, Jeong Geung Lee, Calin Hojbota, Naser Ahmadiniaz, Bobbili Rao, Seong Ku Lee, Jeong Uk Shin, et al.

Constantin Aniculaesei, Thanh Ha, Edward McCarry, Samuel Yoffe, Scott V. Ludedtke, Enrico Brunetti, Michael E. Donovan, Michael Downer, Bernhard Ersfeld, Jose A. Franco-Altamirano, Andrea Hannasch, Dino A. Jaroszynski, Lance Labun, Isabella Pagano, Hernan Quevedo, Ritwik Sain, Michael Spinks, Rafal Zgadzaj, Bjorn Manuel Hegelich

- Collaborators

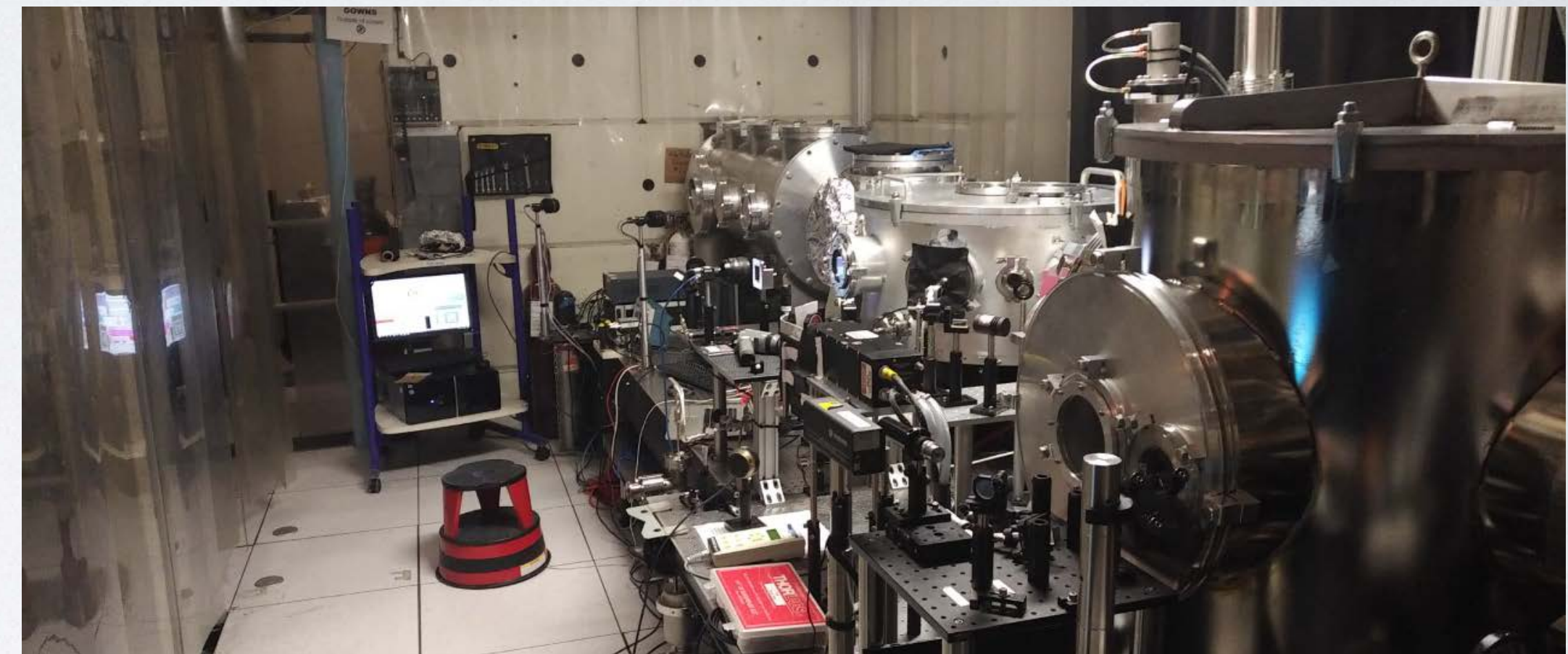
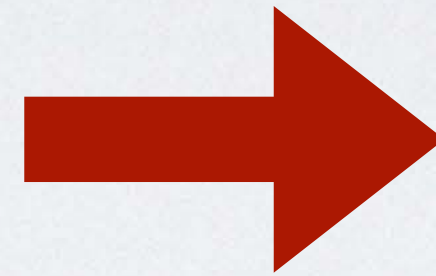
- H. Ruhl; Ludwig-Maximilians-Universität München, Germany
- A. Caldwell, P. Muggli - MPI for Physics, Munich
- E. Gschwendtner et al. - CERN
- J. Zuegel, D. Froula, H. Rinderknecht, M. Campbell - Laboratory for Laser Energetics, Univ. of Rochester
- N. Lemos, F. Albert; Lawrence Livermore National Laboratory
- L. Yin, B. Albright, E. Mottola, Los Alamos National Laboratory
- S. Fleming; University of Arizona, Tucson

- Sponsors

- Air Force Office of Scientific Research
- US Department of Energy
- Institute for Basic Science, Republic of Korea



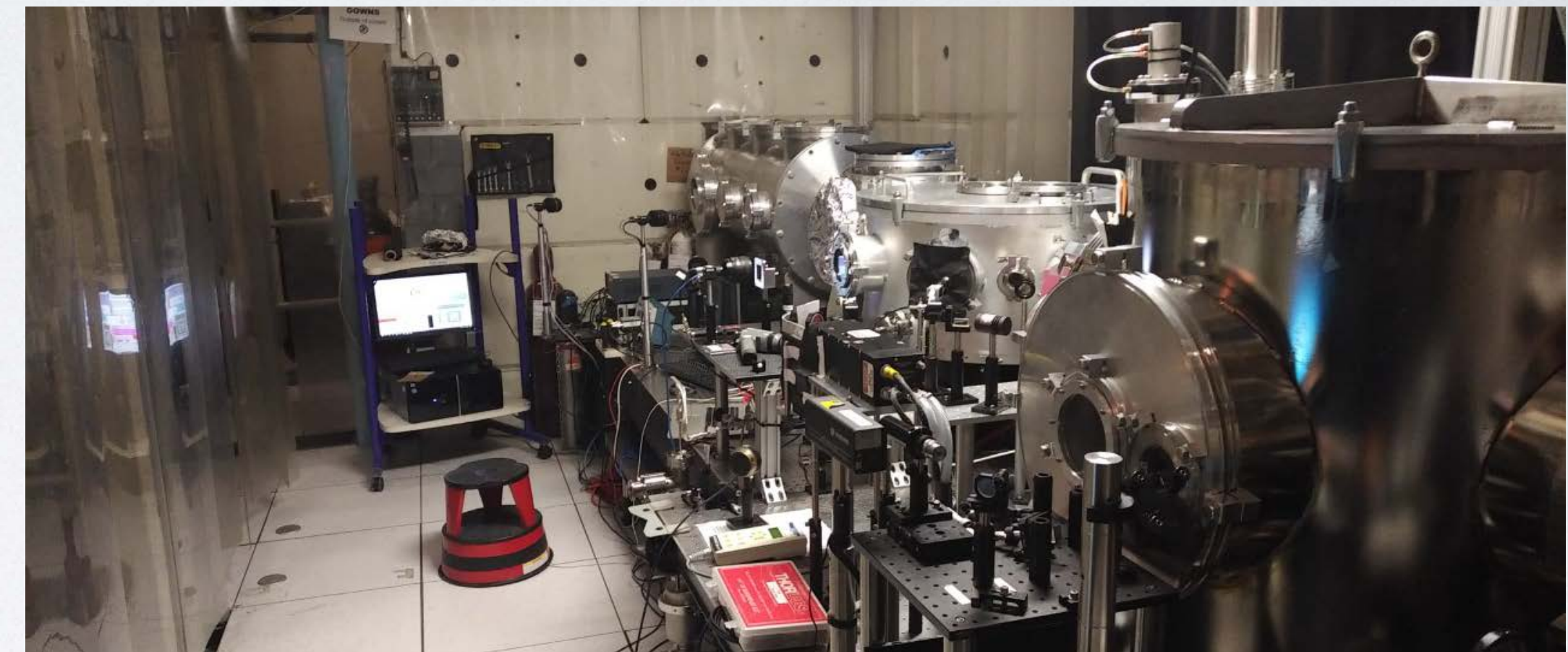
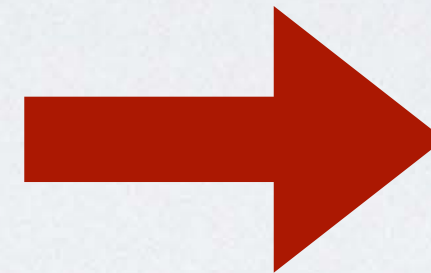
LASER WAKEFIELD ACCELERATION (LWFA)



SHRINKS ACCELERATORS FROM $\sim$ KM TO $\sim$ CM ($\sim$ M)

Compact accelerators and laser-driven XFELs

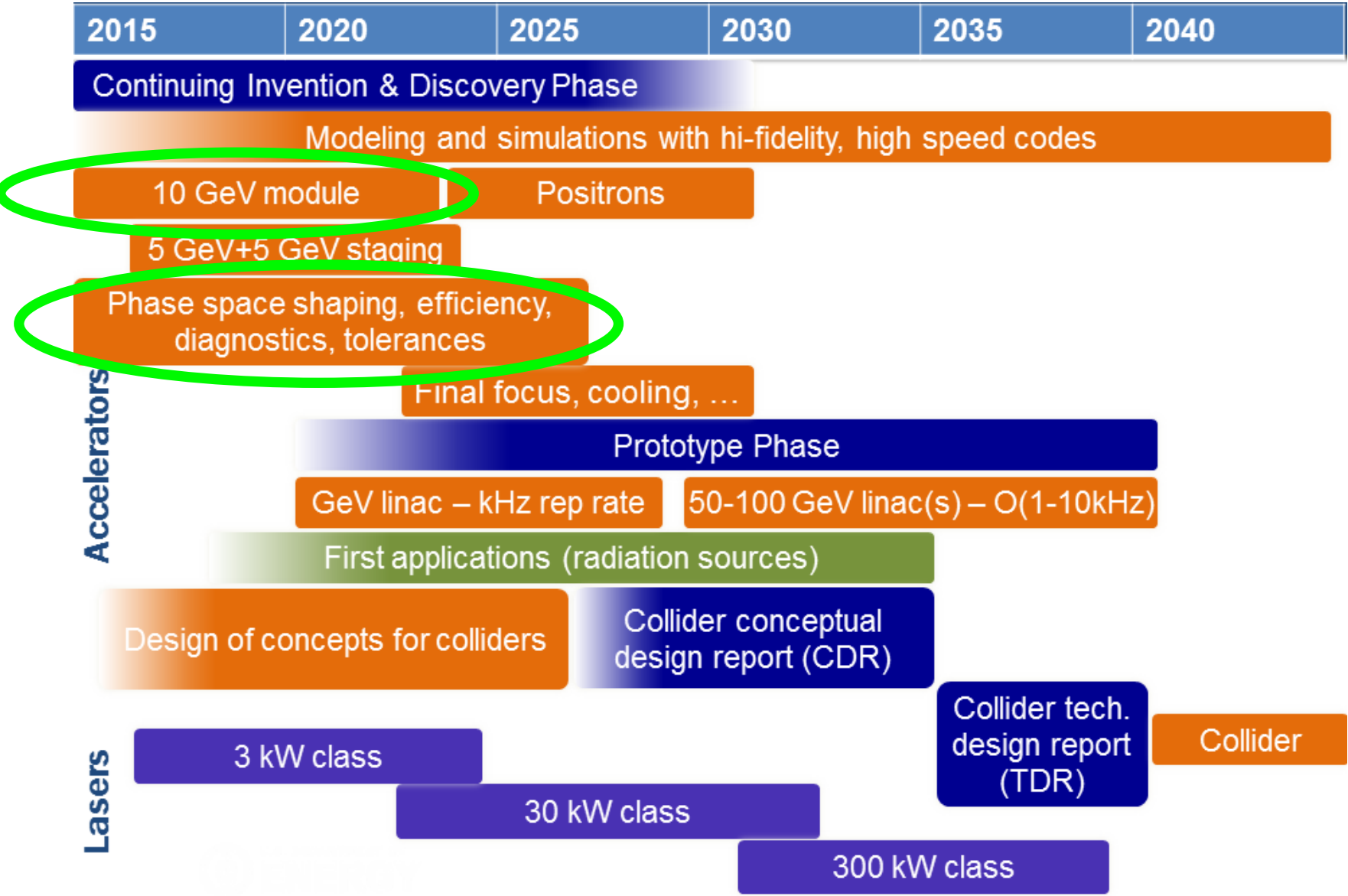
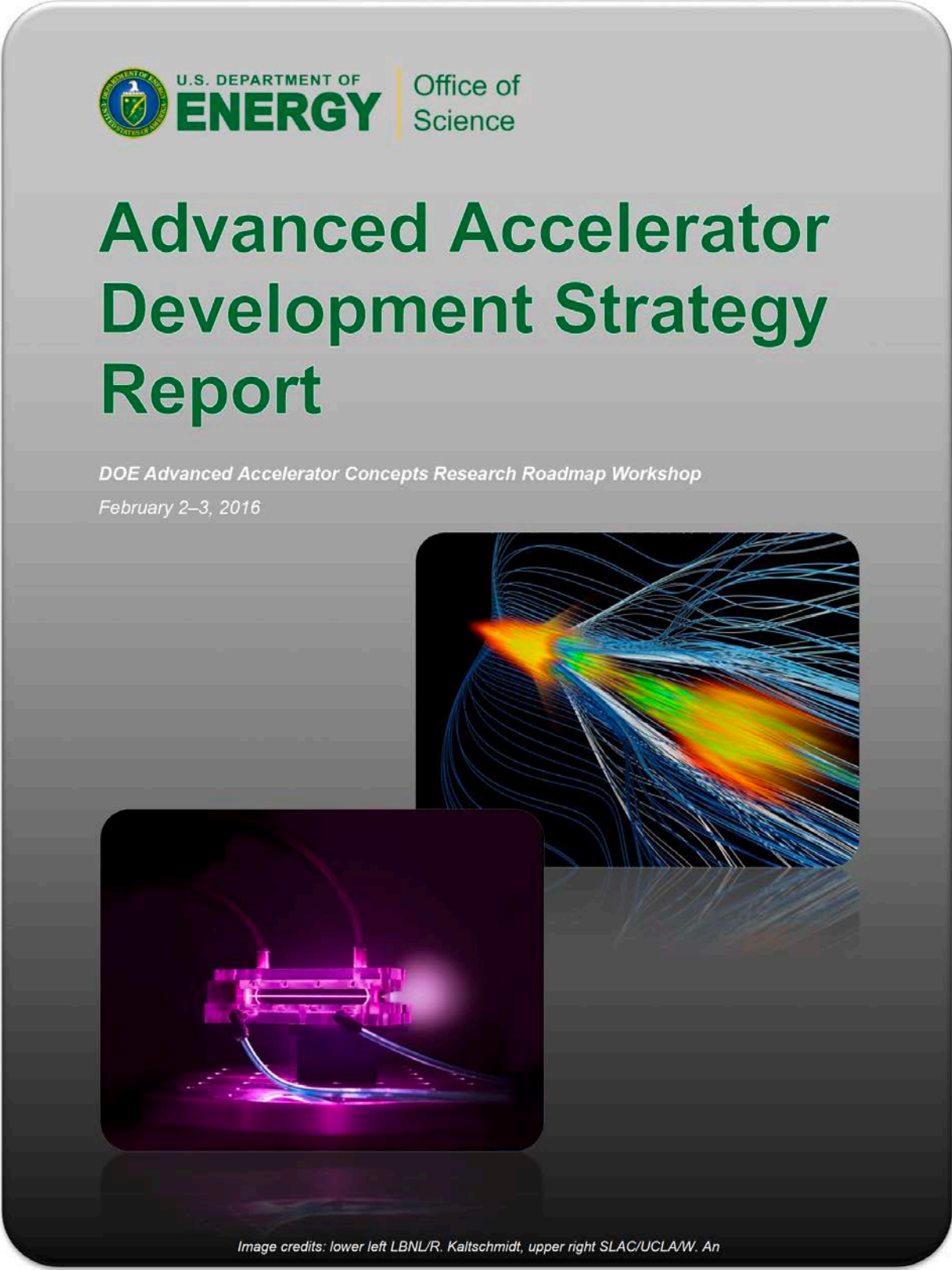
LASER WAKEFIELD ACCELERATION (LWFA)



SHRINKS ACCELERATORS FROM $\sim$ KM TO $\sim$ CM ($\sim$ M)

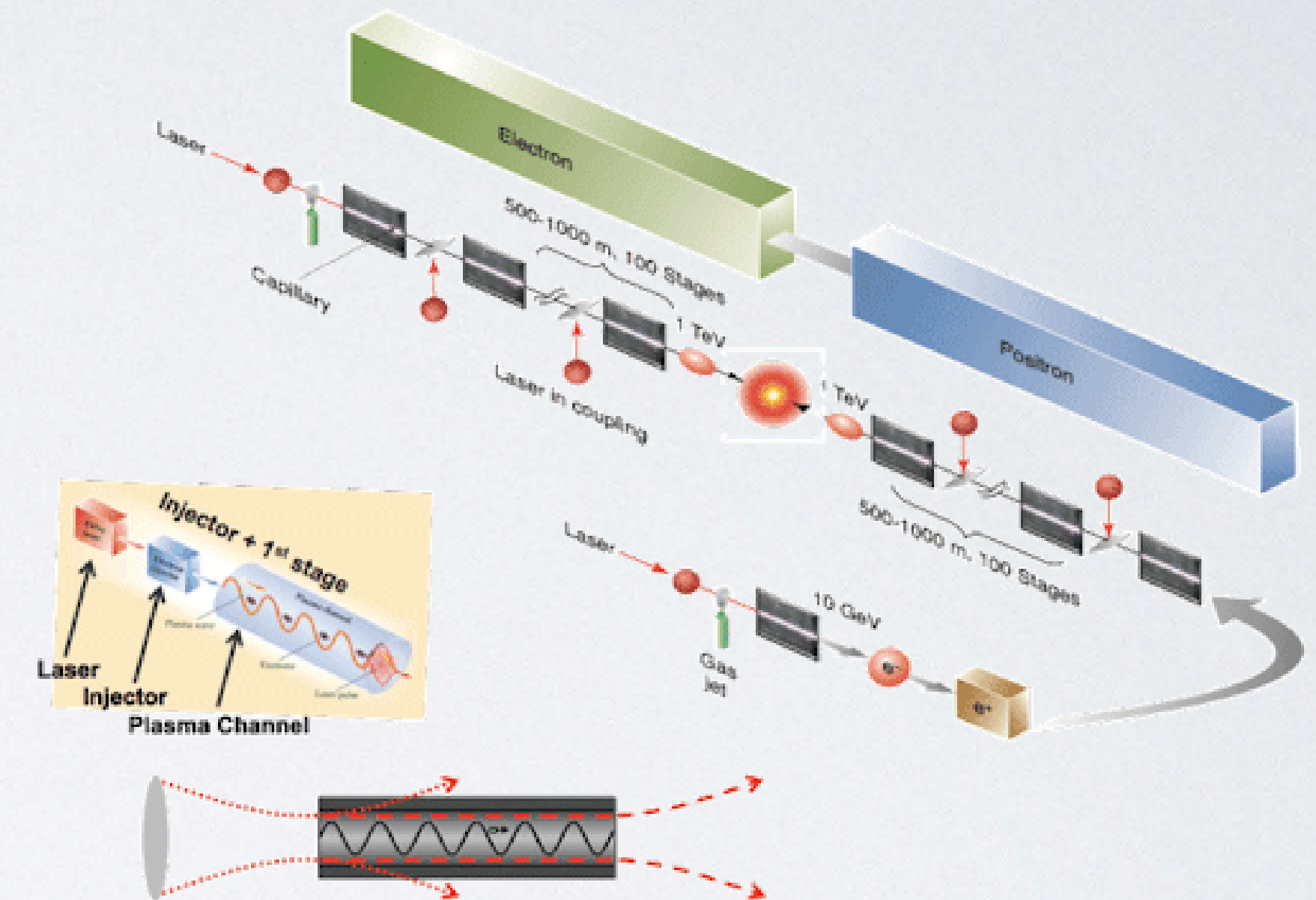
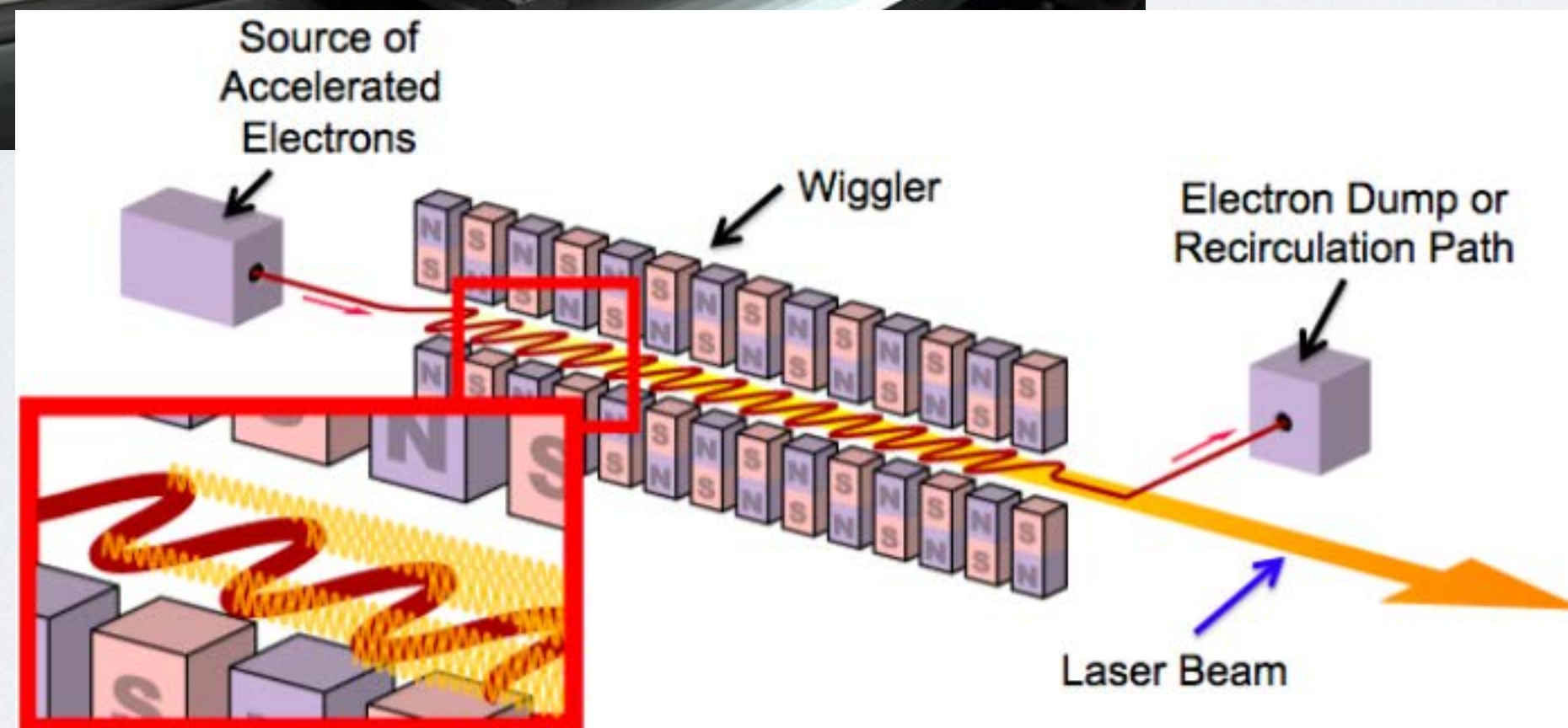
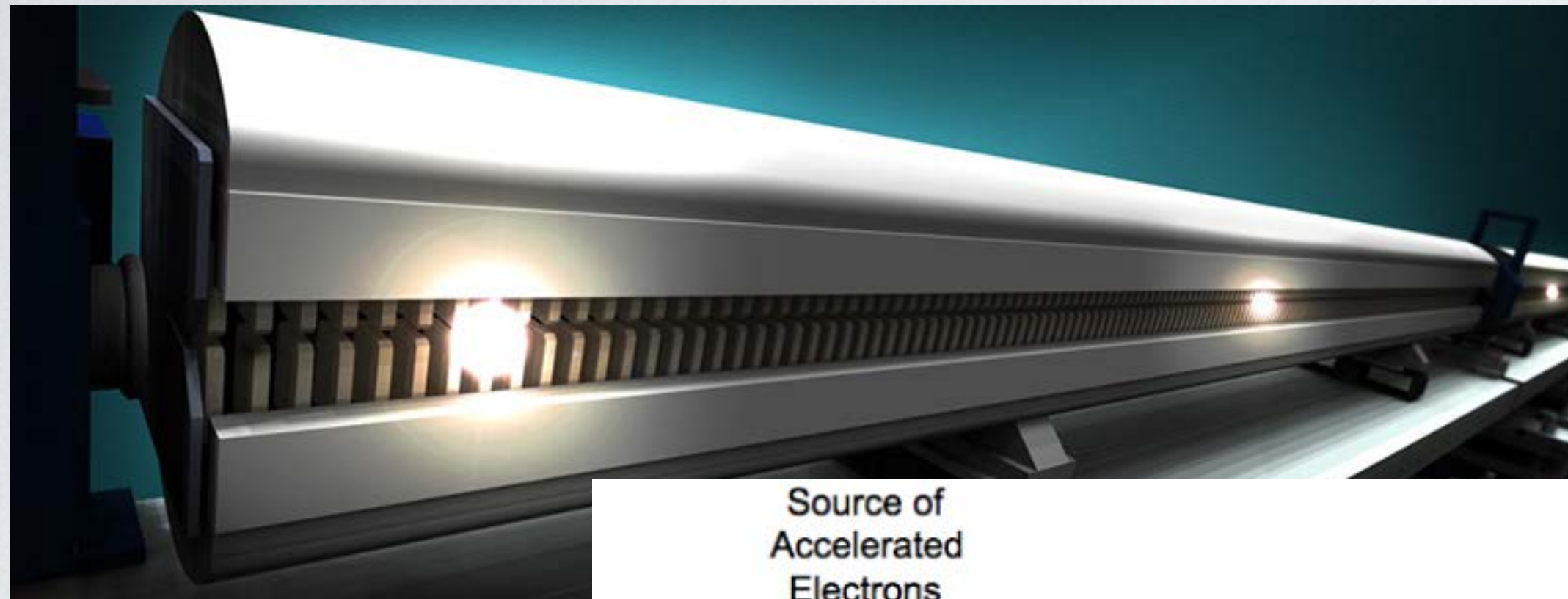
Compact accelerators and laser-driven XFELs

Community Milestone: 10 GeV from a single stage



2016	2018	2020	2022	2024	2026
10 GeV e-beams from a single stage					
Present	Goals	Staging 2.0: demonstration of 5GeV+5GeV			
4.3 GeV	10 GeV	Present	Goals	Positron beams	
30 pC	100 pC	0.1 GeV boost	5 GeV	Goal: novel concept for a compact plasma accelerator based positron source	
Unmatched guiding	Matched guiding	Few pC, 4% captured	100pC, >90% captured	Pair production from LPA generated e-beam	
Fluctuates	Stable, reproducible, tunable	>5 GV/m	>5GV/m	Positron beam captured in PWFA stage	
Second beamline on BELLA		Emittance growth	Emittance preserved	Positron acceleration in laser driven stage	
Laser tech R&D k-BELLA = kW class, kHz, 100 TW laser					
5 Hz, 0.5-1 GeV beam			kHz, 0.5-1 GeV beam		
Present	Goals	Present	Goals		
$\epsilon < 0.3$ micron	$\epsilon < 0.1$ micron	Limited control feedback	Full feedback stabilization		
$\Delta E/E \sim 1\text{-}5\%$	$\Delta E/E < 1\%$	Low average power (<4 W)	High average power (>1 kW)		
Q ~10 pC	Q~10pC	Pointing < 0.5 mrad	Pointing < 0.05 mrad		
γ -ray source (>10 ⁷ ph/s)			γ -ray source (>10 ¹⁰ ph/s)		
LWFA powered FEL (XUV)			LWFA powered FEL (1-10 nm)		
Plasma target and energy recovery technology					
Present	Goals	Goals			
Longitudinally uniform	Tapered	Heat mitigation and >10 ⁸ shots lifetime at kHz			
Parabolic	Near hollow	Photon acceleration to reach high efficiency			
10 cm	>30 cm	Spent laser energy recovery			
1 kHz rep rate	10 kHz rep rate				
Diagnostics					
Goals					
Non-invasive phase space diagnostics for 0.01-0.1-mm-mrad					
Femtosecond resolution for slice properties					
3-D plasma profile vs time					
Simulations					
Present	Goals				
1 D MHD	3 D MHD				
2 weeks for 1 high res 3D BELLA simulation run	<1 Hr for 1 high res 3D BELLA simulation run				

GRAND CHALLENGE: 10GeV from single laser-stage

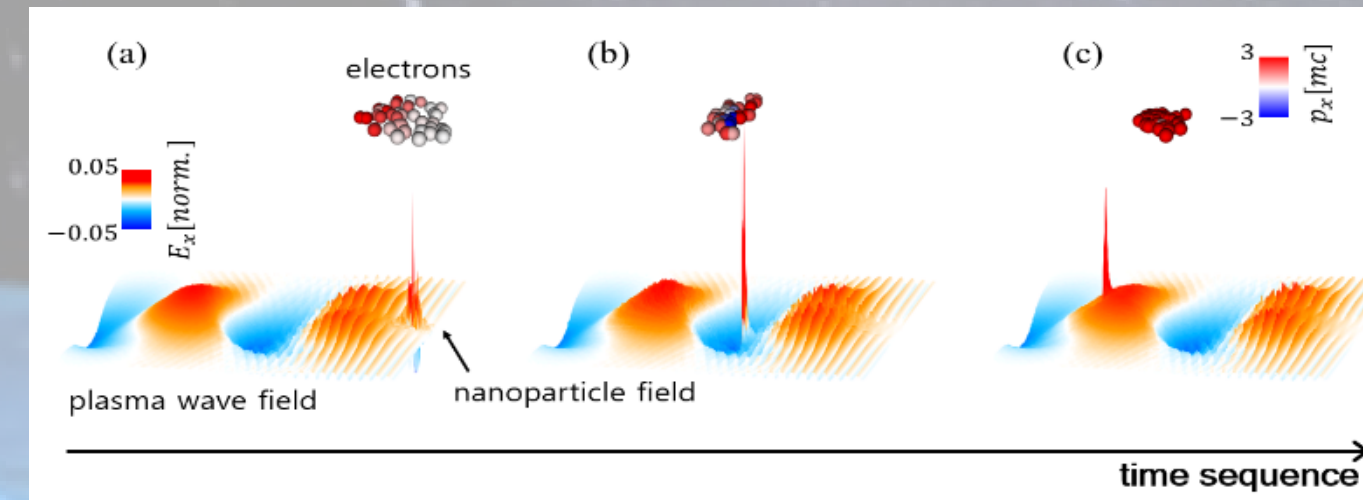


enables room-sized XFEL and affordable TeV collider
(necessary but not sufficient!)

Issues in LWFA research

► Electron injection

- density gradient
- ionization injection

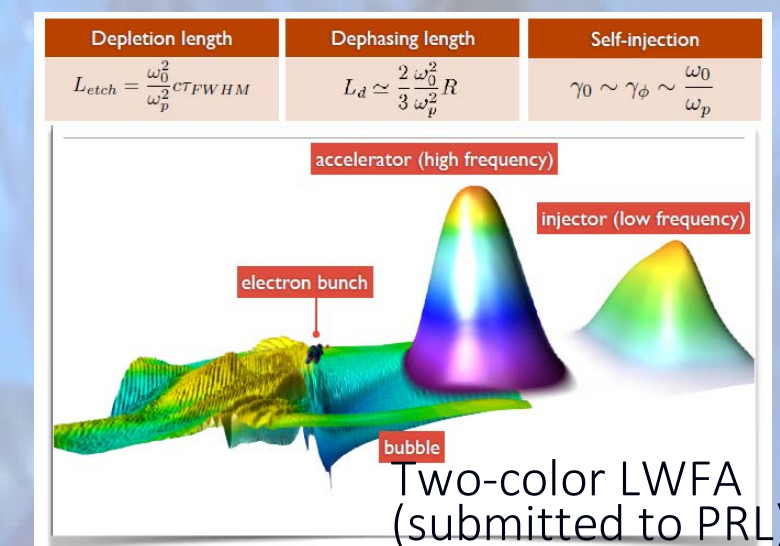


► Laser propagation

- Relativistic self-focusing & channeling
- Self steepening, pulse erosion, ...

► Control of acceleration process

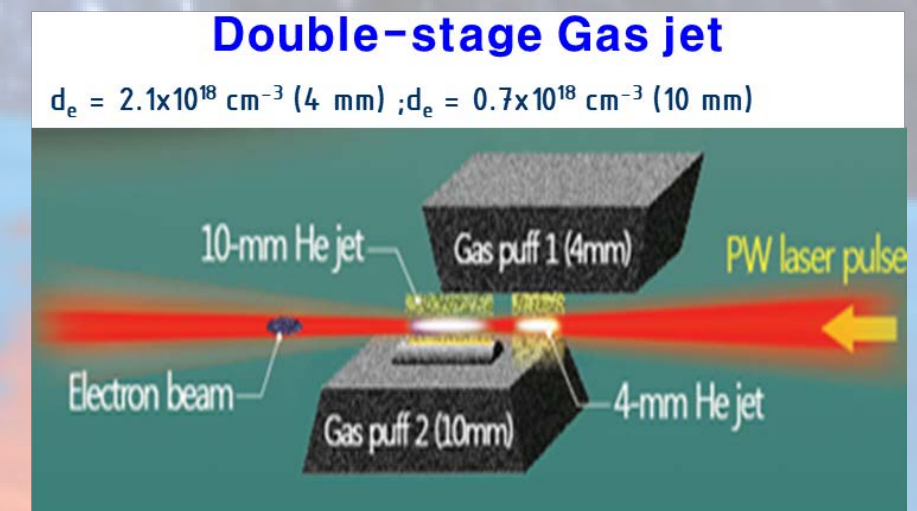
- Dephasing & pump depletion
- Acceleration gradient
- Beam loading effect



► Diagnostics on plasma dynamics

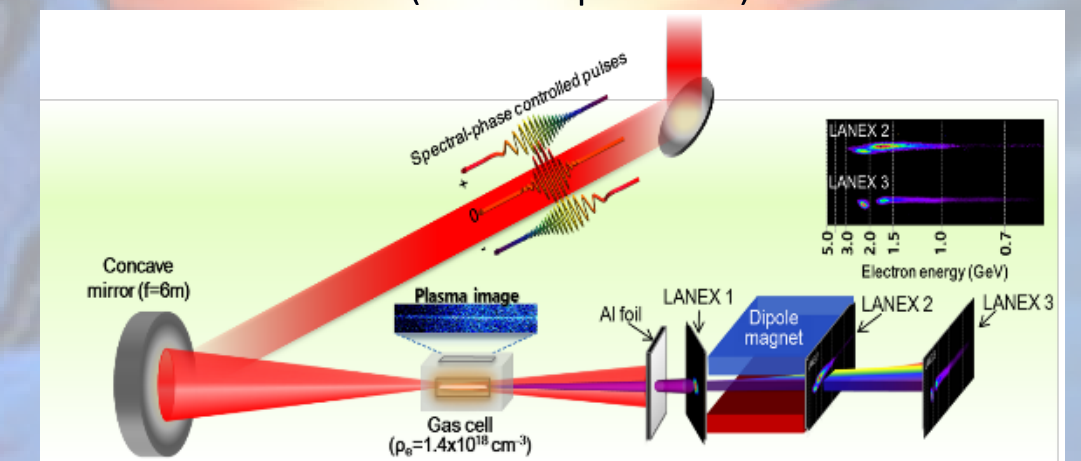
► Radiation sources

- Betatron radiation
- Inverse Compton scattering

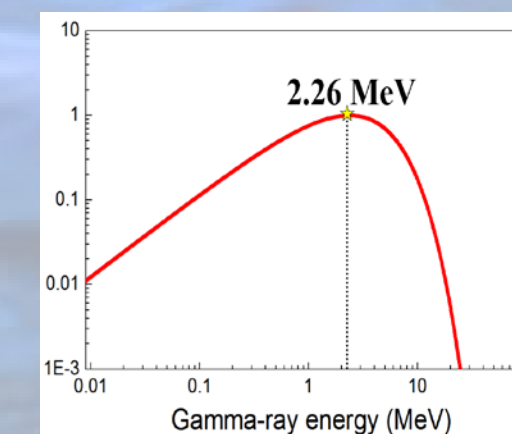


PRL (2013)

Waveform control
(Sci. Rep 2017)



Betatron Rad.
RSI 2015, POP2016



Current record LWFA energy: LBNL - Bella-laser:
7.8 GeV (2019)

RF accelerator bunch injection into LWFA accelerator -
Tsinghua Univ., Y. Wu et al. (2020)

PHYSICAL REVIEW LETTERS 122, 084801 (2019)

Editors' Suggestion Featured in Physics

Petawatt Laser Guiding and Electron Beam Acceleration to 8 GeV
in a Laser-Heated Capillary Discharge Waveguide

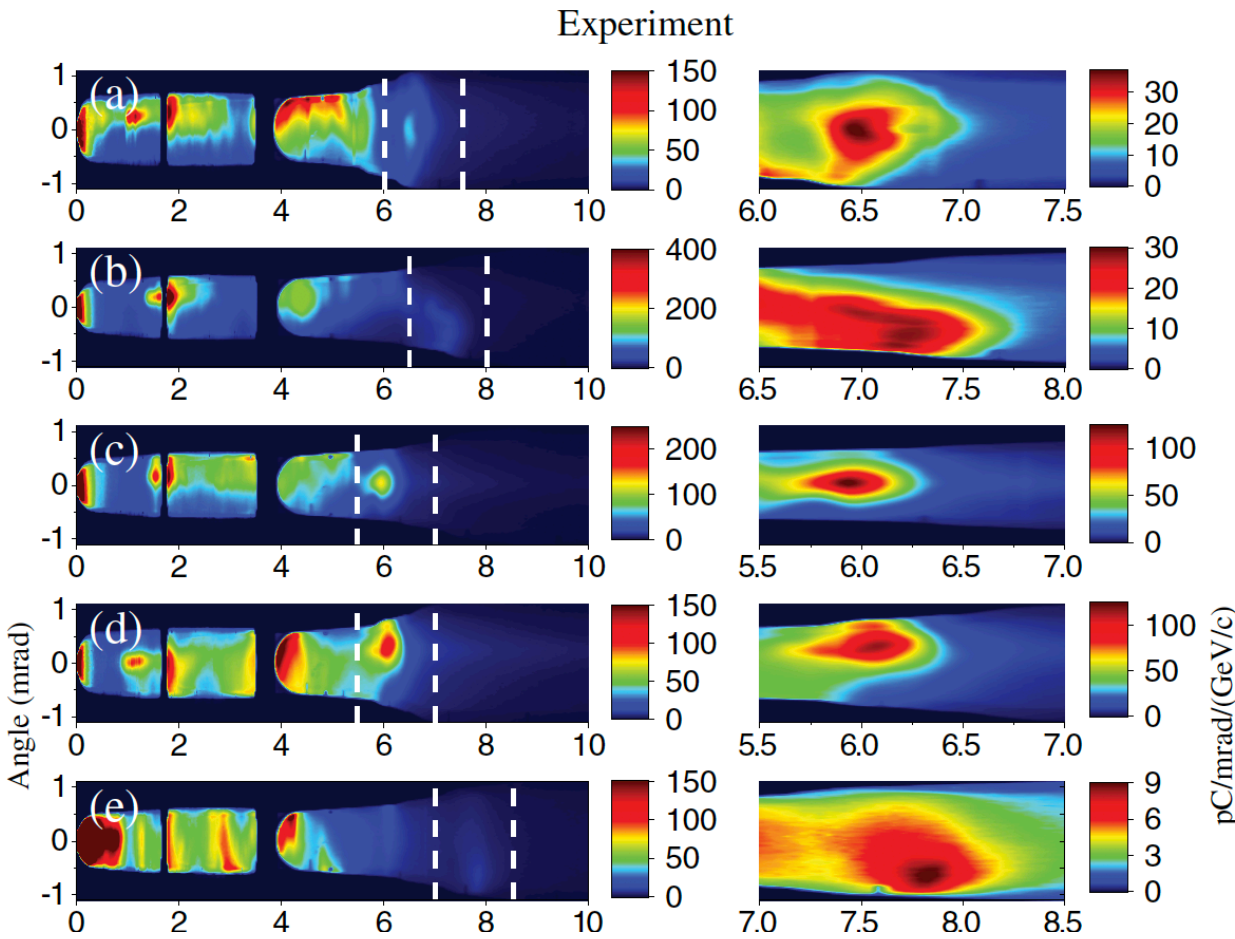
A. J. Gonsalves,^{1,*} K. Nakamura,¹ J. Daniels,¹ C. Benedetti,¹ C. Pieronek,^{1,2} T. C. H. de Raadt,¹ S. Steinke,¹ J. H. Bin,¹ S. S. Bulanov,¹ J. van Tilborg,¹ C. G. R. Geddes,¹ C. B. Schroeder,^{1,2} Cs. Tóth,¹ E. Esarey,¹ K. Swanson,^{1,2} L. Fan-Chiang,^{1,2} G. Bagdasarov,^{3,4} N. Bobrova,^{3,5} V. Gasilov,^{3,4} G. Korn,⁶ P. Sasorov,^{3,6} and W. P. Leemans^{1,2,†}

¹Lawrence Berkeley National Laboratory, Berkeley, California 94720, USA
²University of California, Berkeley, California 94720, USA
³Keldysh Institute of Applied Mathematics RAS, Moscow 125047, Russia
⁴National Research Nuclear University MEPhI (Moscow Engineering Physics Institute), Moscow 115409, Russia
⁵Faculty of Nuclear Science and Physical Engineering, CTU in Prague, Břehova 7, Prague 1, Czech Republic
⁶Institute of Physics ASCR, v.v.i. (FZU), ELI-Beamlines Project, 182 21 Prague, Czech Republic

(Received 7 December 2018; revised manuscript received 30 January 2019; published 25 February 2019)

Guiding of relativistically intense laser pulses with peak power of 0.85 PW over 15 diffraction lengths was demonstrated by increasing the focusing strength of a capillary discharge waveguide using laser inverse bremsstrahlung heating. This allowed for the production of electron beams with quasimonochromatic peaks up to 7.8 GeV, double the energy that was previously demonstrated. Charge was **5 pC at 7.8 GeV** and up to **62 pC in 6 GeV peaks**, and typical beam divergence was 0.2 mrad.

DOI: 10.1103/PhysRevLett.122.084801



nature
physics

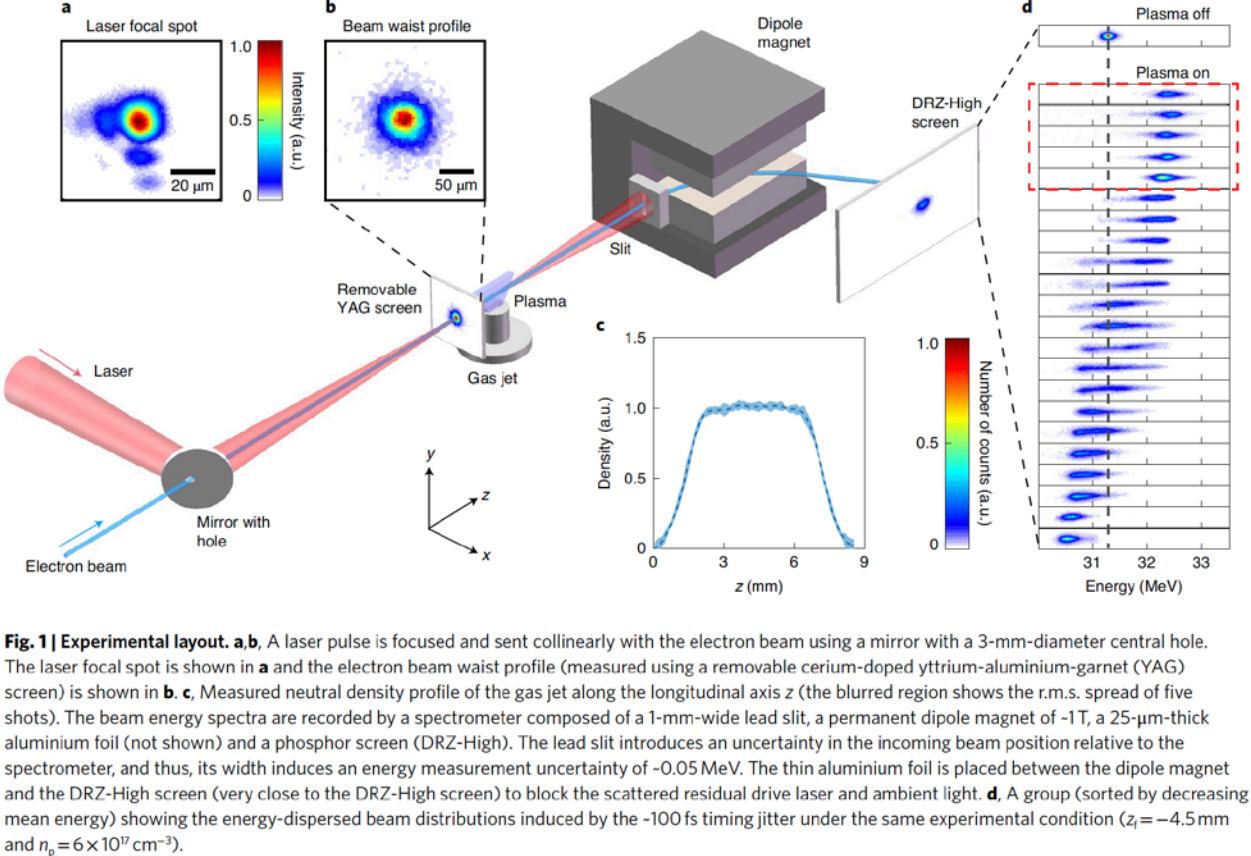
LETTERS

https://doi.org/10.1038/s41567-021-01202-6

Check for updates

High-throughput injection-acceleration of
electron bunches from a linear accelerator to a
laser wakefield accelerator

Yipeng Wu^{1,2}, Jianfei Hua^{1,3}, Zheng Zhou¹, Jie Zhang¹, Shuang Liu¹, Bo Peng¹, Yu Fang¹, Xiaonan Ning¹, Zan Nie², Fei Li², Chaojie Zhang², Chih-Hao Pai¹, Yingchao Du^{1,3}, Wei Lu^{1,3}, Warren B. Mori² and Chan Joshi²



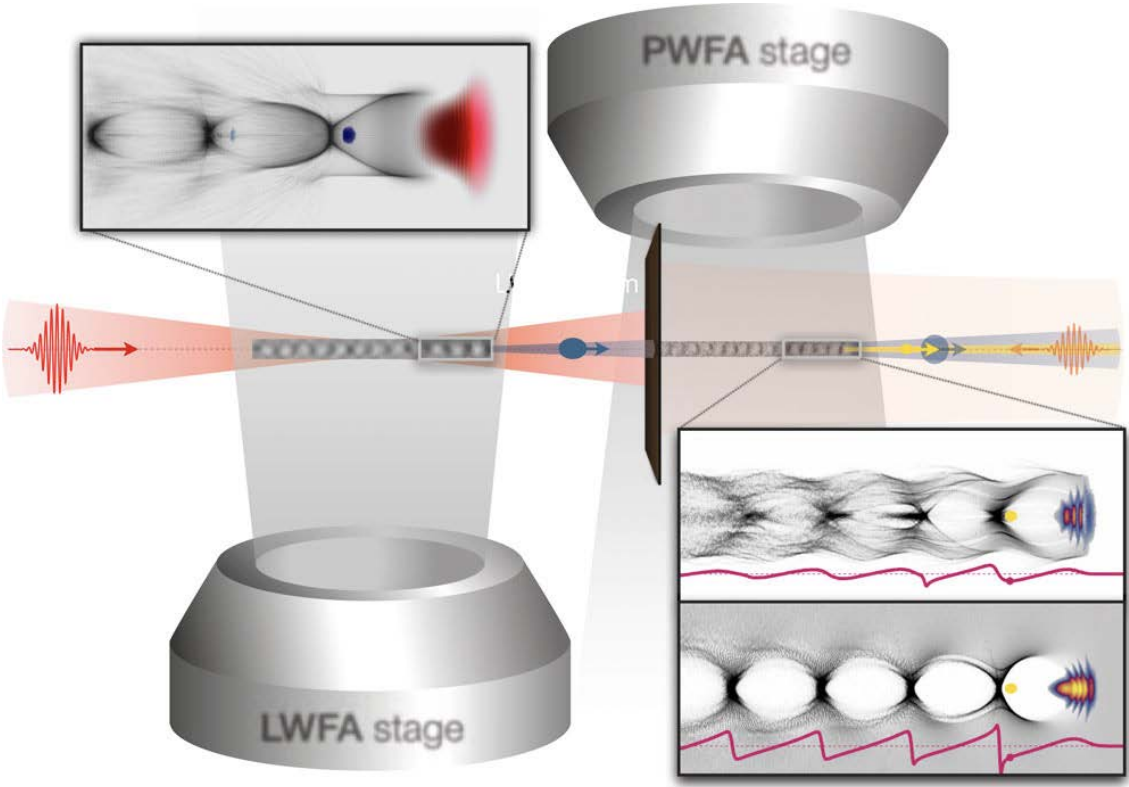
LWFA powered PWFA - Kurz et al., HZDR (2020)

ARTICLE

https://doi.org/10.1038/s41467-021-23000-7 OPEN

Demonstration of a compact plasma accelerator
powered by laser-accelerated electron beams

T. Kurz^{1,2,10}, T. Heinemann^{3,4,5,10}, M. F. Gilljohann^{6,7}, Y. Y. Chang¹, J. P. Couperus Cabadağ¹, A. Debus¹, O. Kononenko⁸, R. Pausch¹, S. Schöbel^{1,2}, R. W. Assmann³, M. Bussmann^{1,9}, H. Ding^{6,7}, J. Götzfried^{6,7}, A. Köhler¹, G. Raj⁸, S. Schindler^{6,7}, K. Steiniger¹, O. Zarini¹, S. Corde⁸, A. Döpp^{6,7}, B. Hidding^{4,5}, S. Karsch^{6,7}, U. Schramm^{1,2}, A. Martinez de la Ossa³ & A. Irman¹



Longterm stable operation of an LWFA using Bayesian
Optimization and Machine Learning - Desy (2021)

PHYSICAL REVIEW LETTERS 126, 104801 (2021)

Bayesian Optimization of a Laser-Plasma Accelerator

Sören Jalas^{1,*}, Manuel Kirchen¹, Philipp Messner^{2,1,3}, Paul Winkler^{3,1}, Lars Hübner^{3,1}, Julian Dirkwinkel³, Matthias Schnepf¹, Remi Lehe⁴ and Andreas R. Maier^{3,1}

¹Center for Free-Electron Laser Science and Department of Physics Universität Hamburg, Luruper Chaussee 149, 22761 Hamburg, Germany
²International Max Planck Research School for Ultrafast Imaging & Structural Dynamics, Luruper Chaussee 149, 22761 Hamburg, Germany
³Deutsches Elektronen Synchrotron (DESY), Notkestraße 85, 22607 Hamburg, Germany
⁴Lawrence Berkeley National Laboratory, Berkeley, California 94720, USA

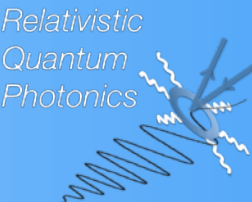
Optimal Beam Loading in a Laser-Plasma Accelerator

Manuel Kirchen^{1,*}, Sören Jalas¹, Philipp Messner^{2,1}, Paul Winkler^{3,1}, Timo Eichner¹, Lars Hübner^{3,1}, Thomas Hülsenbusch^{3,1}, Laurids Jeppe¹, Trupen Parikh³, Matthias Schnepf¹ and Andreas R. Maier^{3,1}

¹Center for Free-Electron Laser Science and Department of Physics Universität Hamburg, Luruper Chaussee 149, 22761 Hamburg, Germany
²International Max Planck Research School for Ultrafast Imaging and Structural Dynamics, Luruper Chaussee 149, 22761 Hamburg, Germany
³Deutsches Elektronen Synchrotron (DESY), Notkestraße 85, 22607 Hamburg, Germany

(Received 11 August 2020; revised 16 December 2020; accepted 2 March 2021; published 26 April 2021)

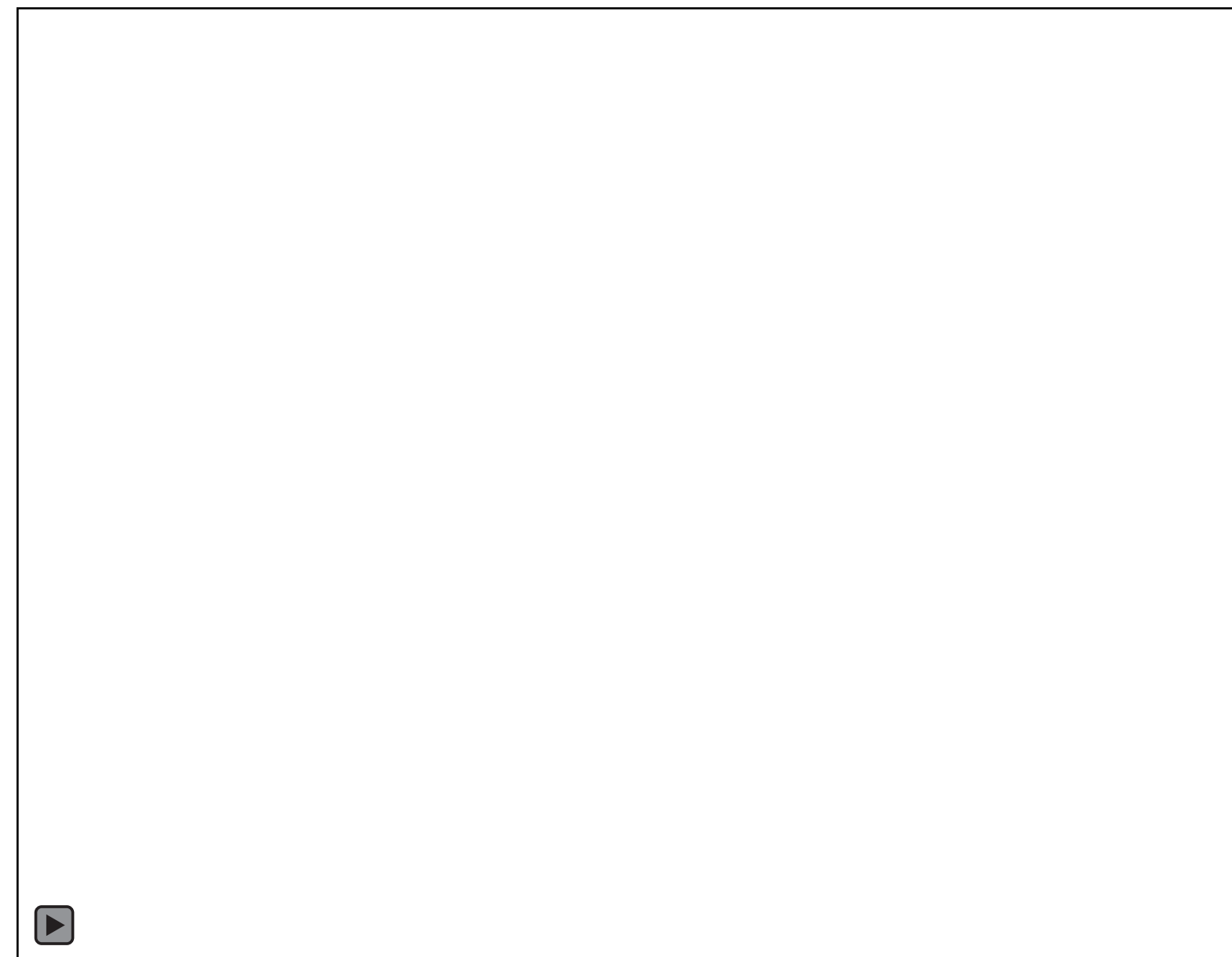
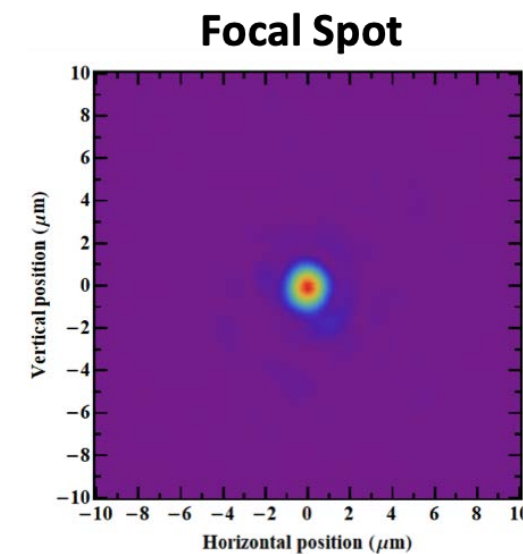
- <1.2% rms energy spread @ 284 MeV, 44pC, by optimal beam loading through ionization injection



The 4 +1 PW laser system at the Center for Relativistic Laser Science in South Korea

4PW Laser System

- Petawatt Laser System
 - 2 beam lines: 4PW + 1.5 PW, Ti:Sa
 - 80J (30J), 20fs, 0.1Hz
 - Intensity up to 10^{23} W/cm²
- 3 experimental stations with permanent setups
 - Long Focus - Wakefield: electrons, betatron, Compton, muons, ...
 - Short Focus - Solid Targets: Ions, neutrons, relativistic HHG, gammas
 - Short - Medium Focus - 2 multi-PW beams: ultrahigh intensity, RR, QED
- First Results in LWFA
 - Electrons up to ~5 GeV
 - Nanoparticle injection: Improved energy, reduced dE/E, divergence



CoReLS laser can in principle produce electron bunches for collision experiment with $\chi > 1$

- Laser parameters :

- 52 J on target, 25fs => 2PW

- focal spot $\approx 50 \mu\text{m}$ (FWHM), + 30 fs (GDD +350 fs²),

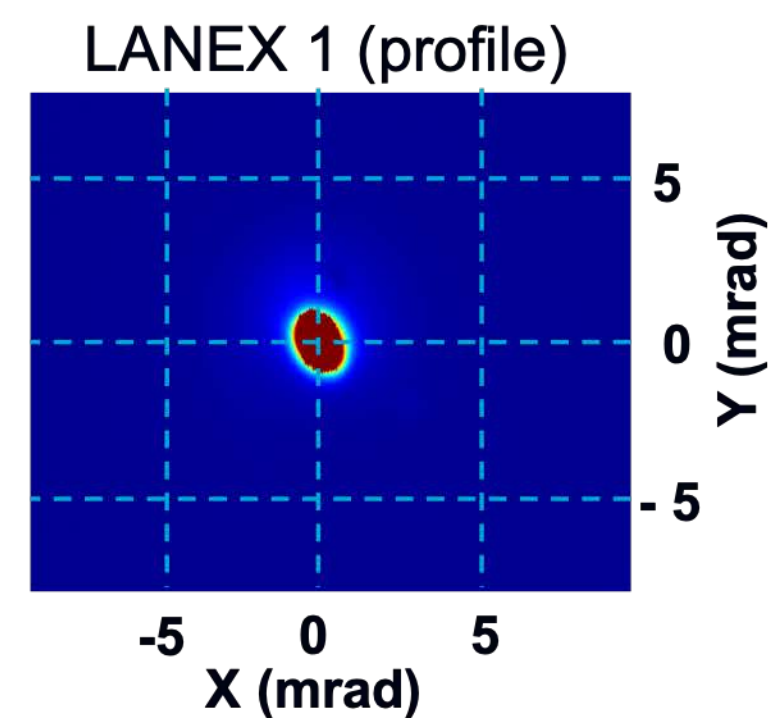
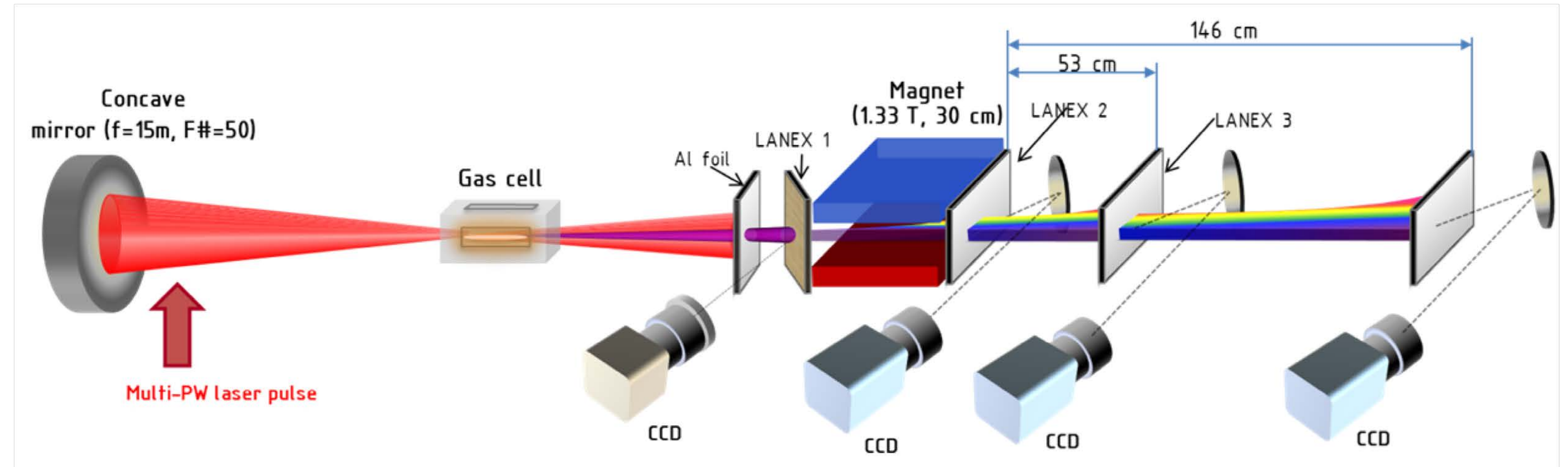
- $I \approx 4.2 \times 10^{19} \text{ W/cm}^2$, $a_0 \approx 4.5$

- Gas medium :

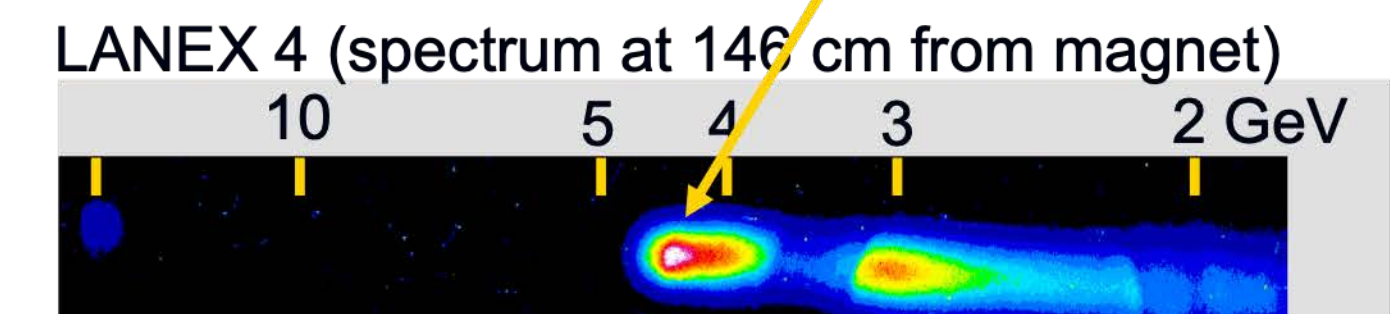
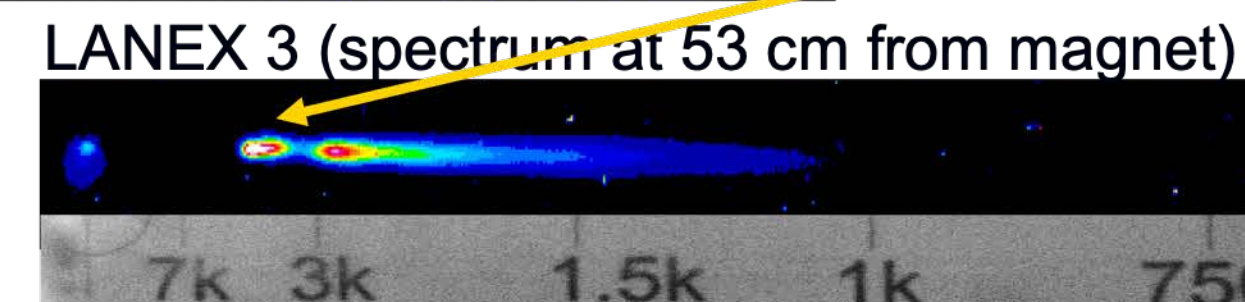
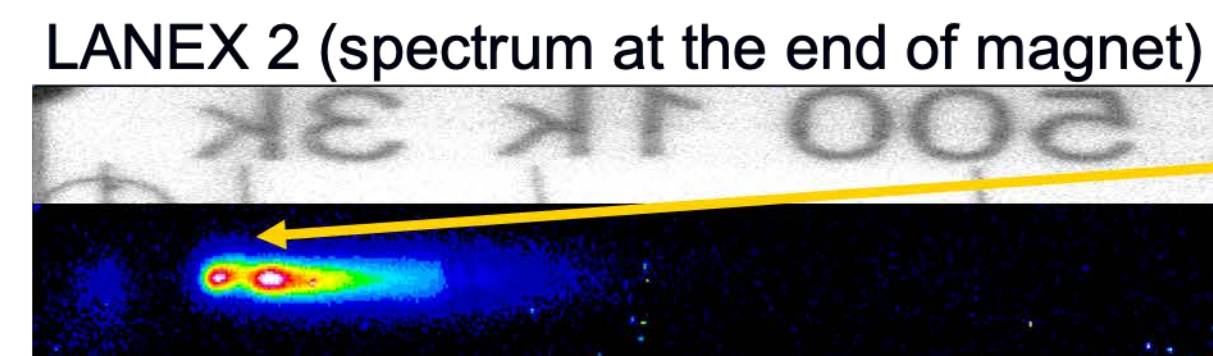
- He mixed with 1-% Ne,

- 7-cm gas cell

- plasma density $\approx 1.5 \times 10^{18} \text{ elec./cc}$



Deeply saturated
Divergence < 2 mrad
Total Charge > 1 nC



Electron energy at peak
 $\approx 4.5 \text{ GeV}$

A signpost with two directional signs. The top sign is tilted and points left, while the bottom sign is horizontal and points right. Both signs are white with black borders and text. The background is a blue sky with white clouds.

TOUTES DIRECTIONS

AUTRES DIRECTIONS

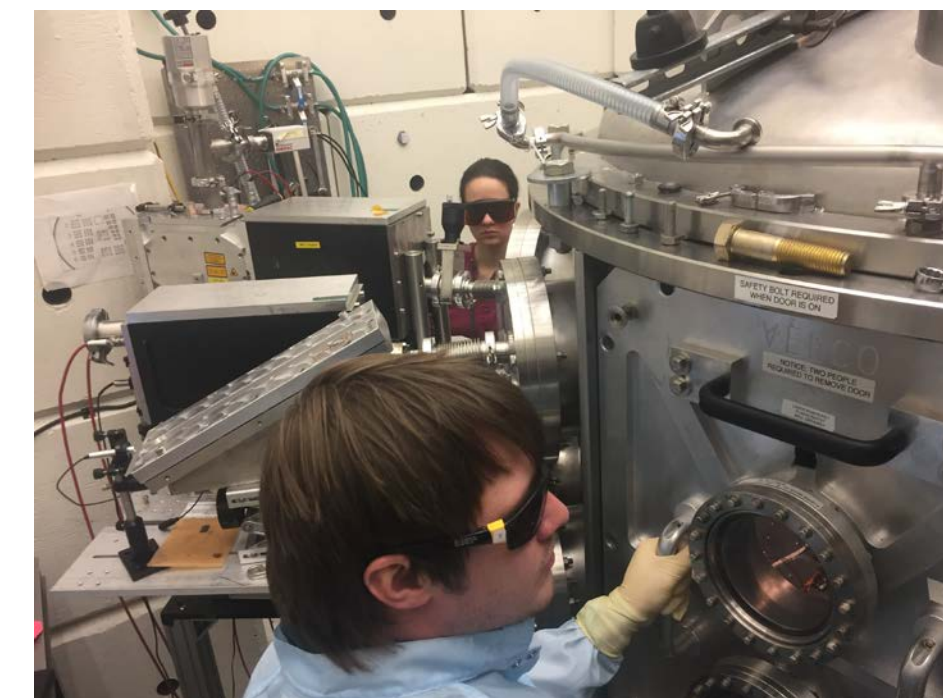
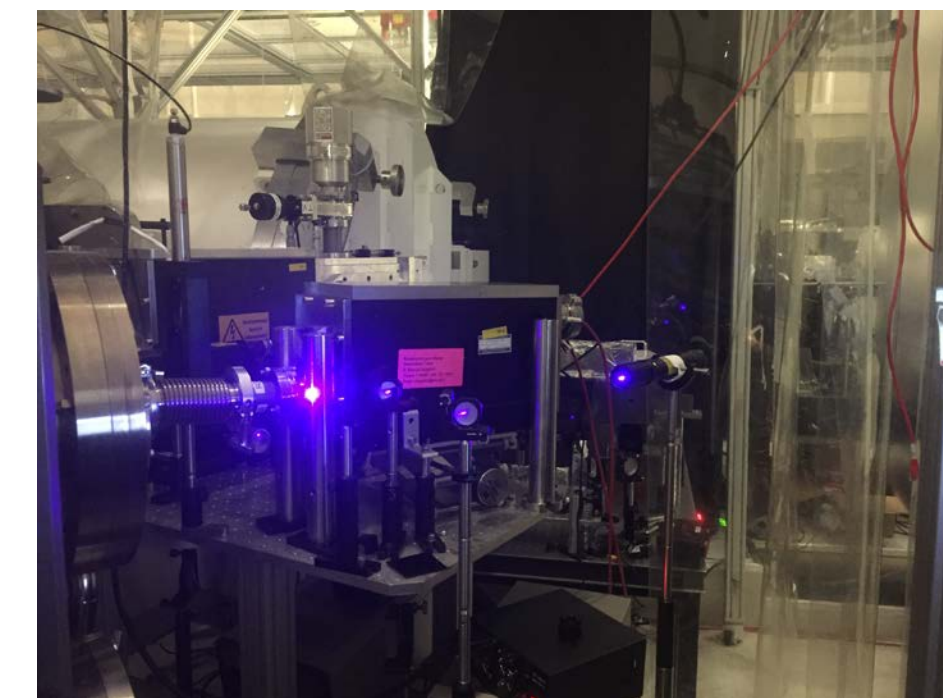
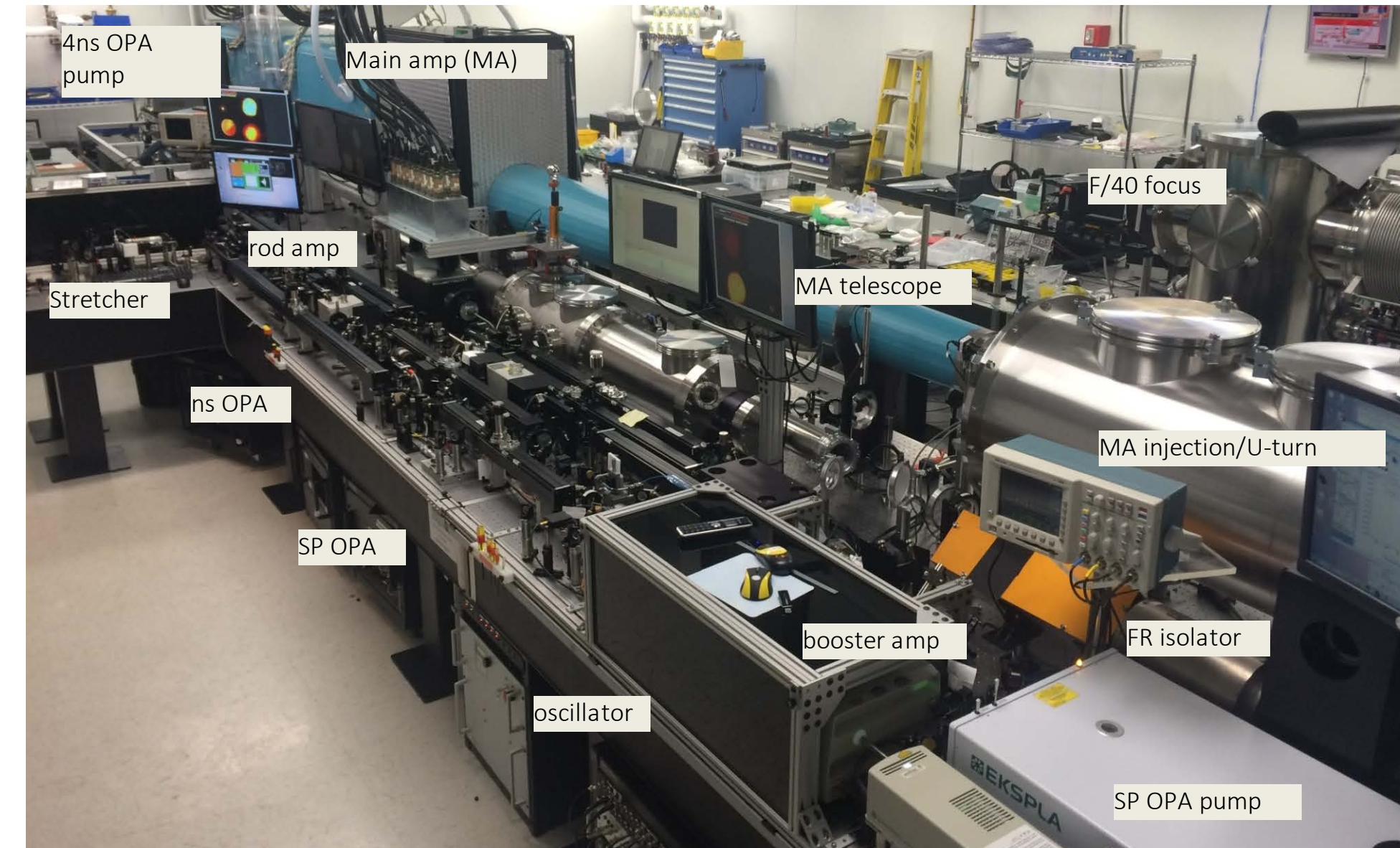
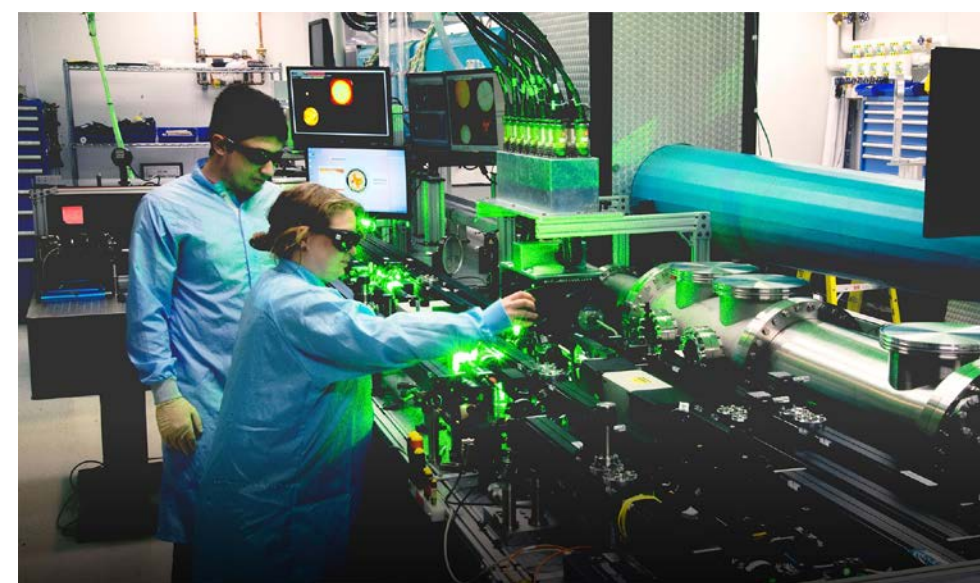
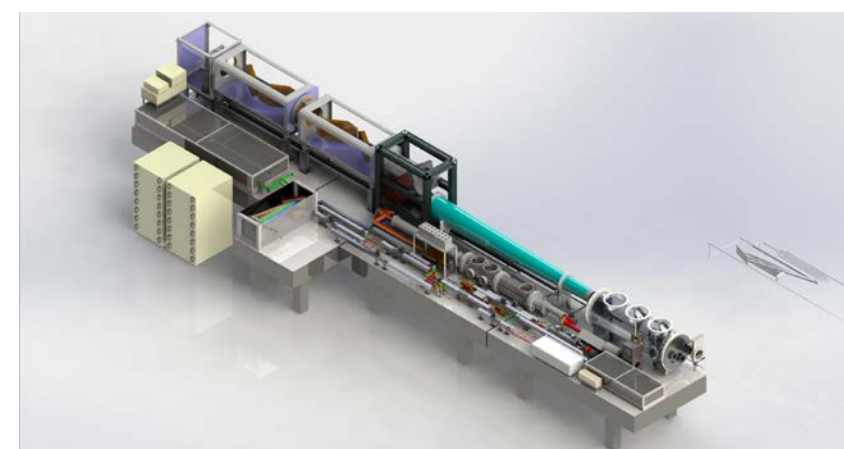
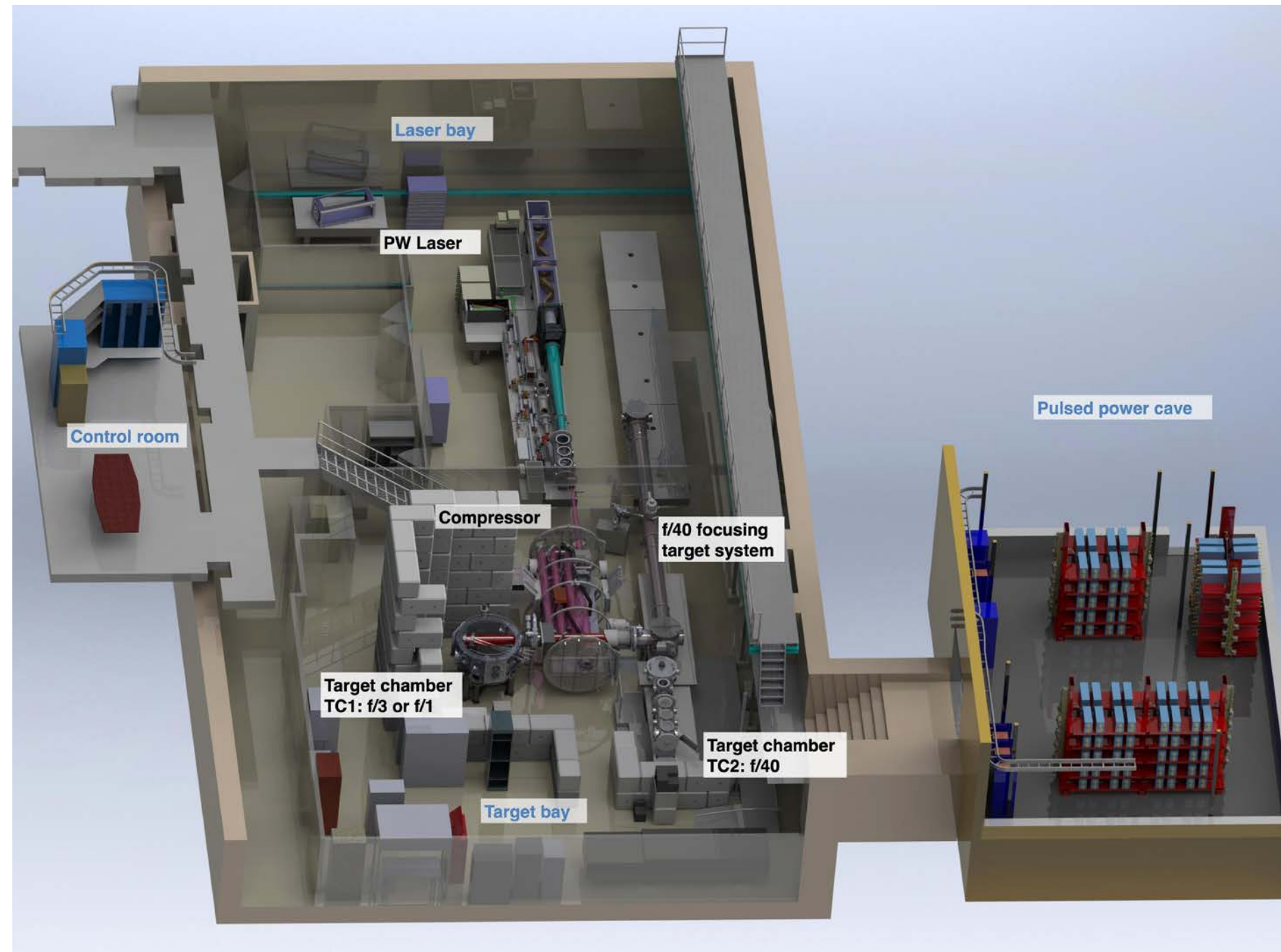
The Texas Petawatt Laser at UT Austin

Laser Parameters:

- 150 J
- 150 fs
- 1 shot/hr
- 2 Target Areas:
 - F/40: 2×10^{18} W/cm²
 - F/3: 10^{21} W/cm²
 - F/1: $> 3 \times 10^{22}$ W/cm²

Demonstrated Performance

- 3 GeV electrons
- 100 MeV protons
- 600 MeV carbon
- 4.4 GeV Au
- $> 10^{10}$ neutron/shot
- > 50 MeV γ -ray beam



Canonical TPW LWFA result: 2012 record

Xiaoming Wang, M. Downer, et al.



ARTICLE

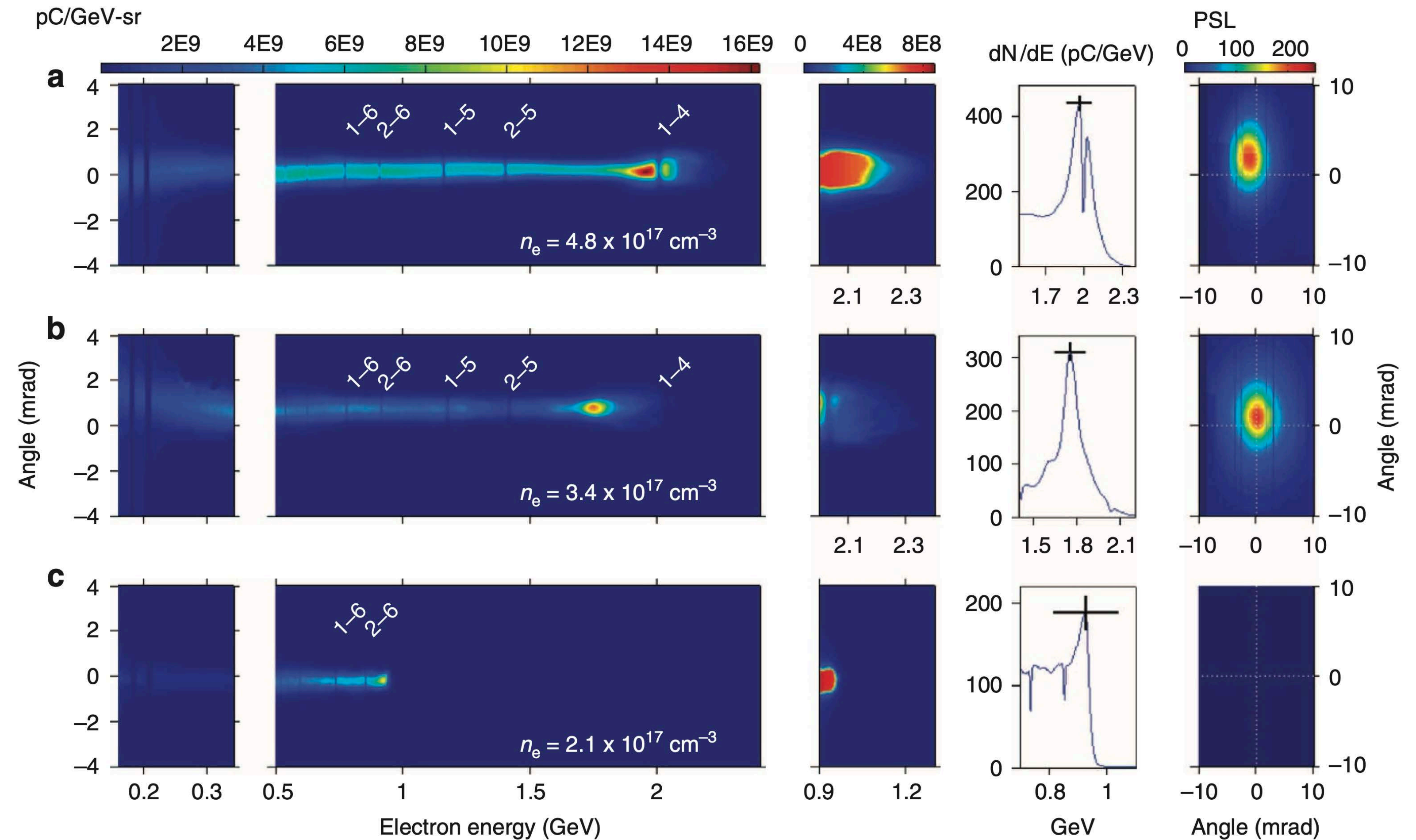
Received 2 Dec 2012 | Accepted 8 May 2013 | Published 11 Jun 2013

DOI: 10.1038/ncomms2988

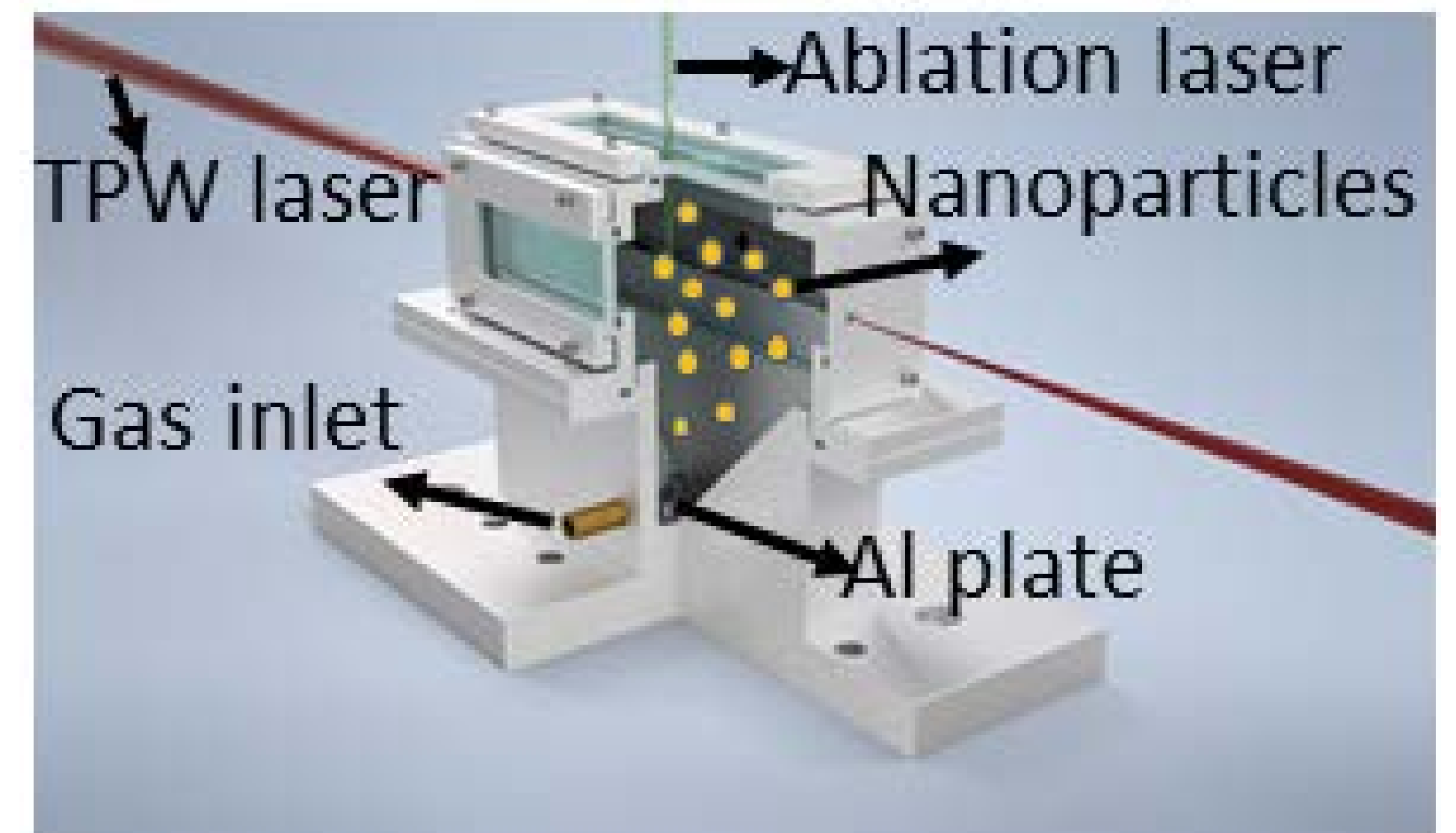
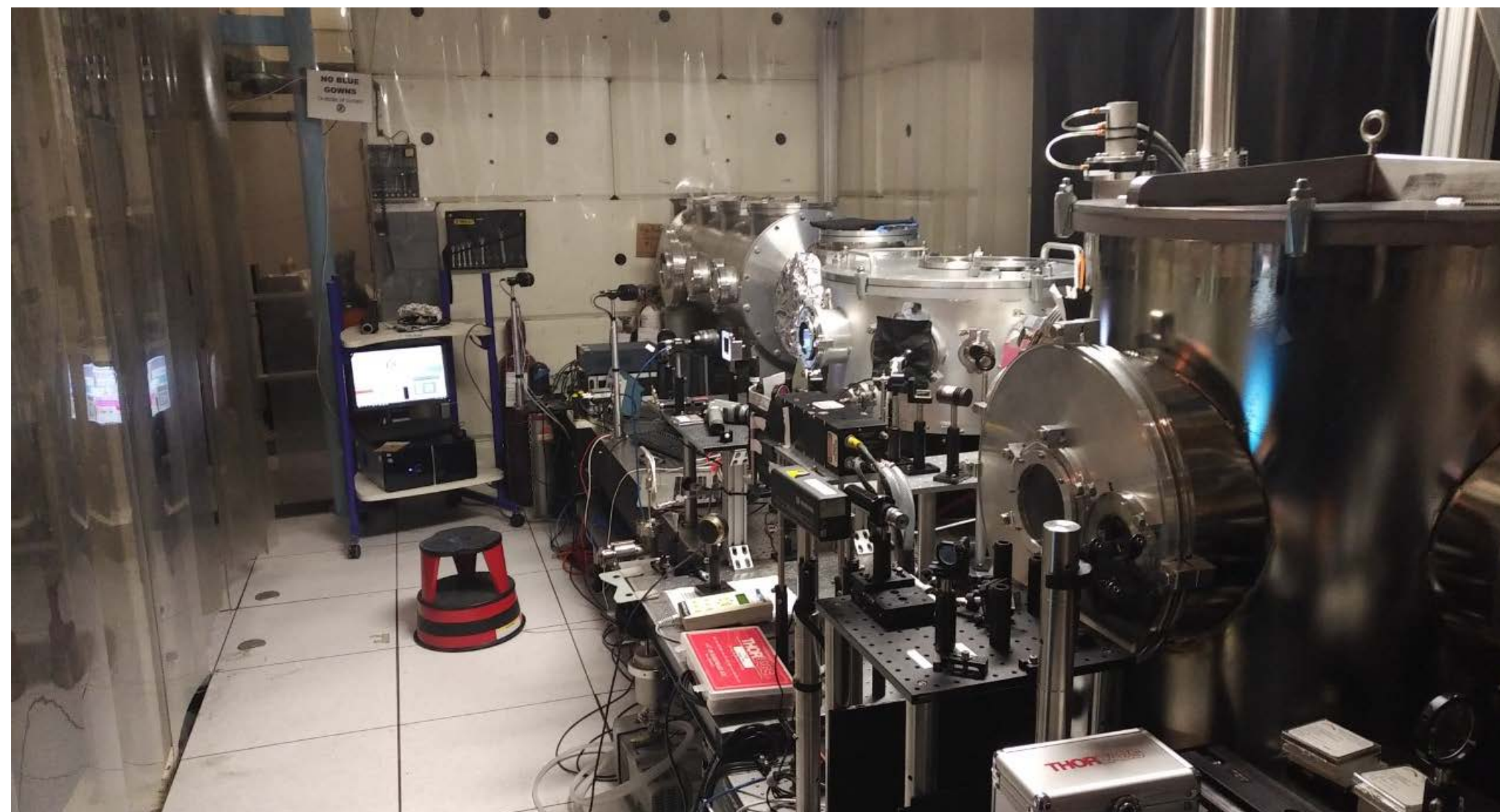
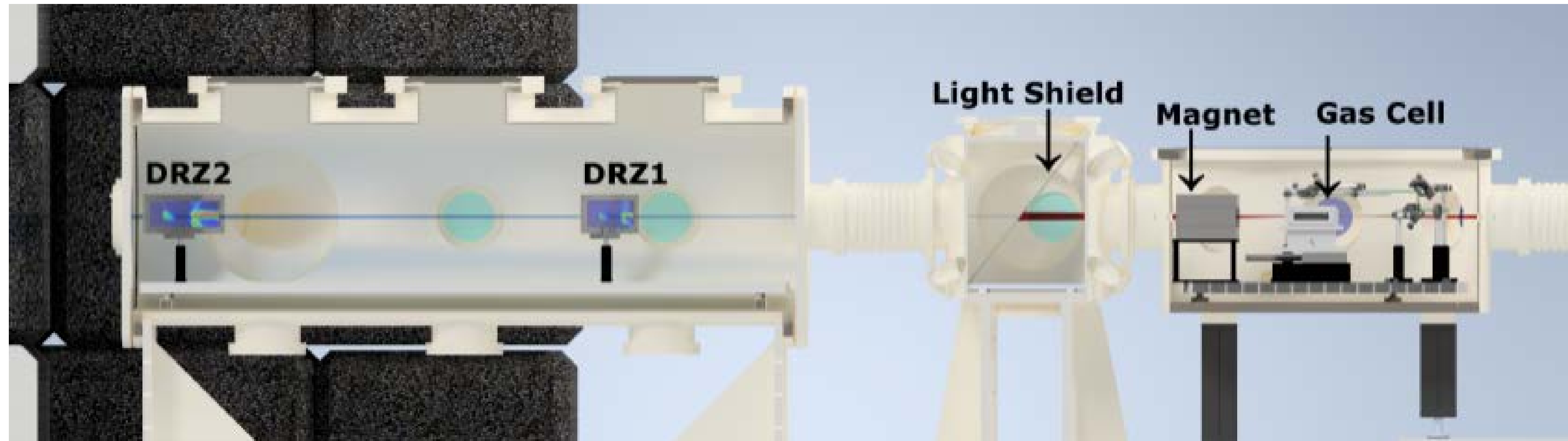
OPEN

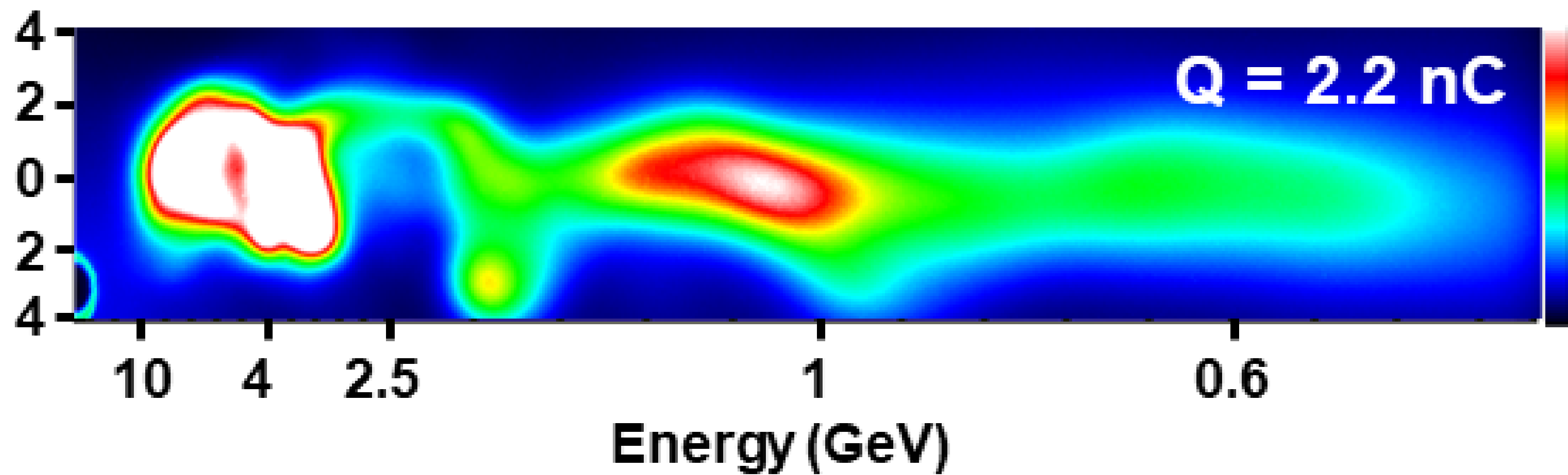
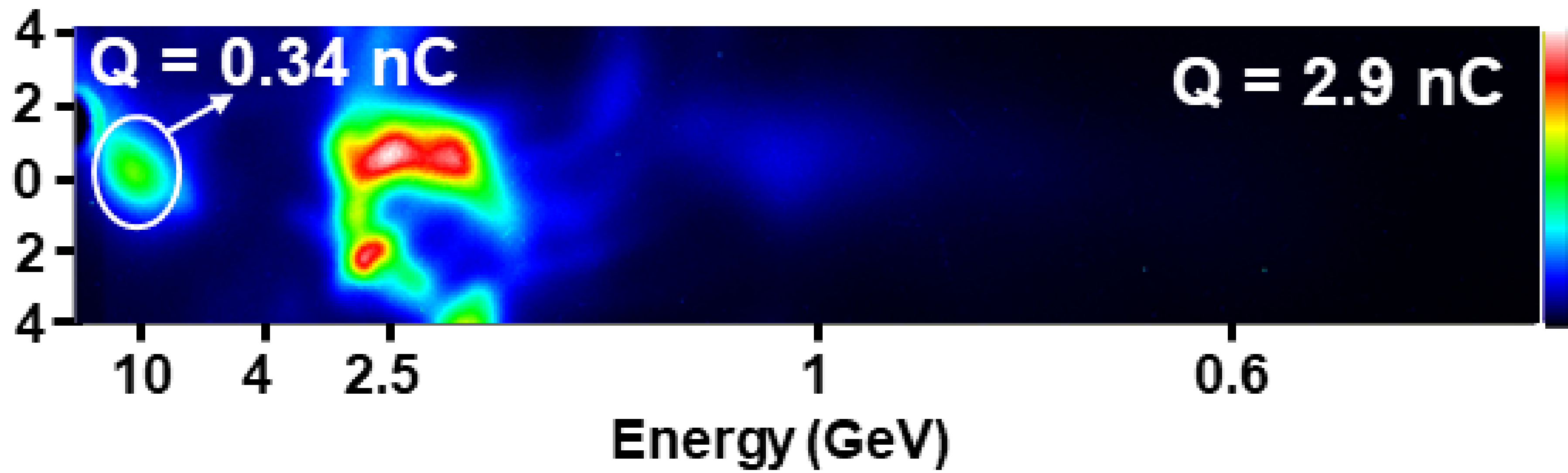
Quasi-monoenergetic laser-plasma acceleration of electrons to 2 GeV

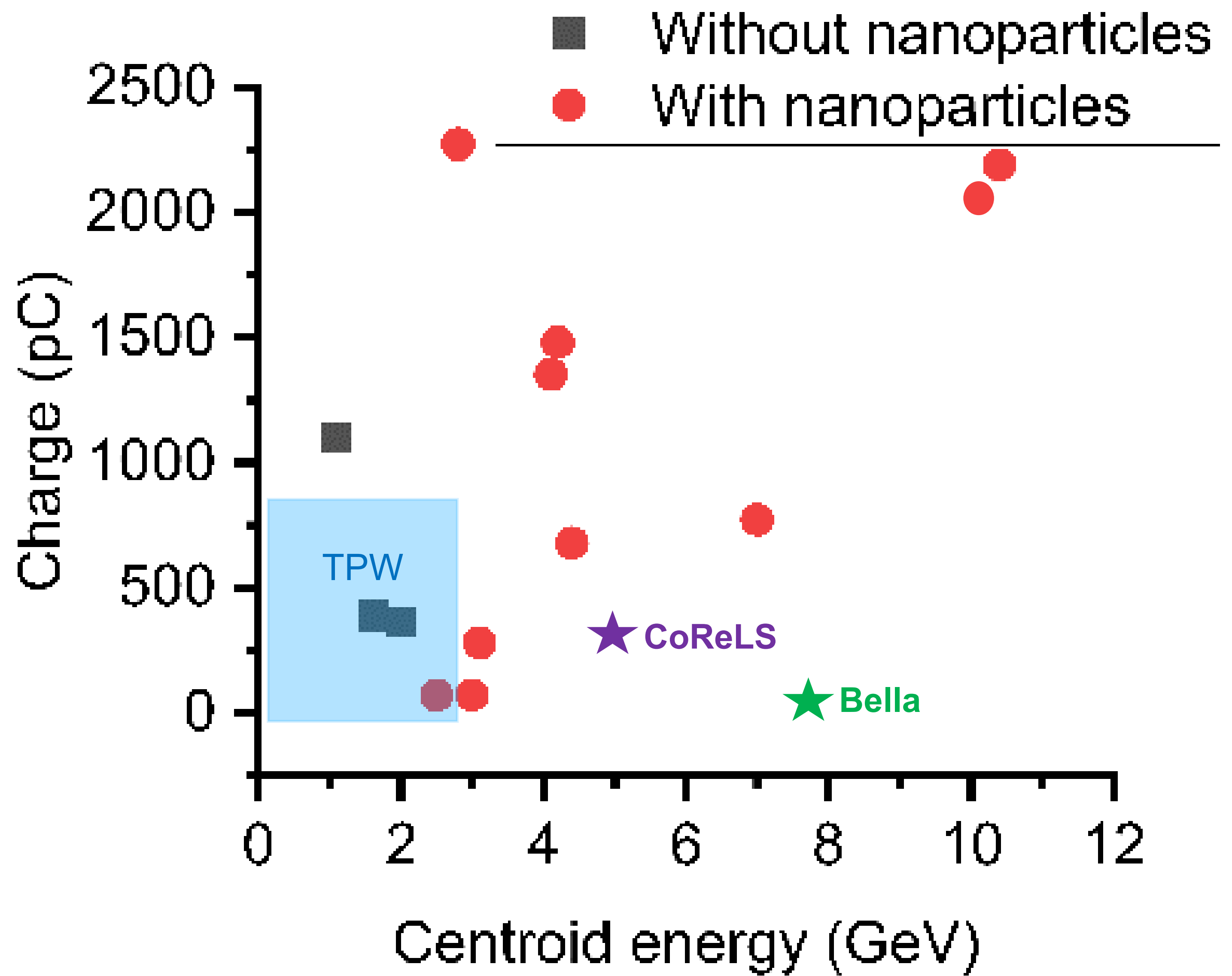
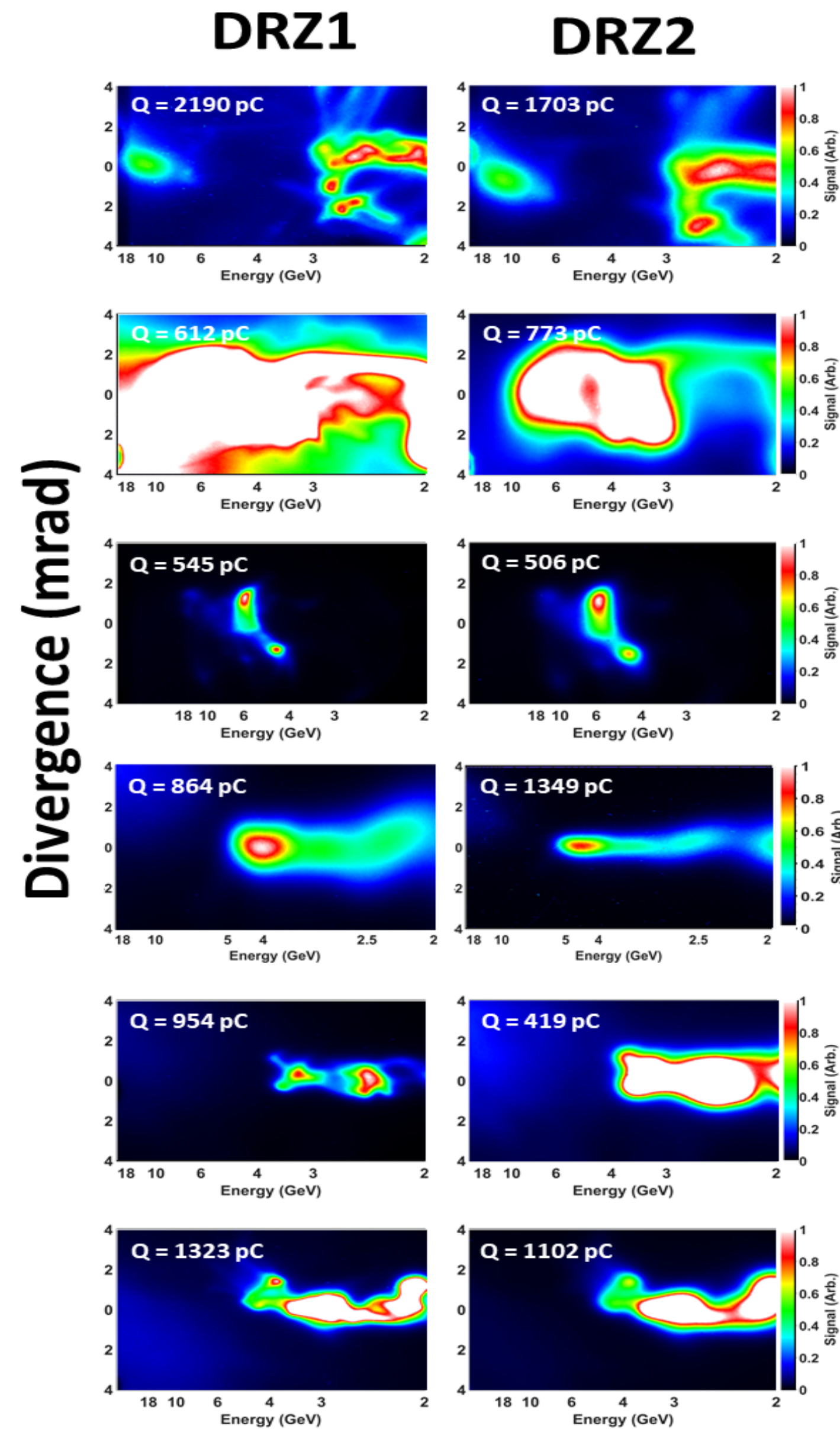
Xiaoming Wang¹, Rafal Zgadzaj¹, Neil Fazel¹, Zhengyan Li¹, S. A. Yi¹, Xi Zhang¹, Watson Henderson¹, Y.-Y. Chang¹, R. Korzekwa¹, H.-E. Tsai¹, C.-H. Pai¹, H. Quevedo¹, G. Dyer¹, E. Gaul¹, M. Martinez¹, A. C. Bernstein¹, T. Borger¹, M. Spinks¹, M. Donovan¹, V. Khudik¹, G. Shvets¹, T. Ditmire¹ & M. C. Downer¹



Experimental Setup







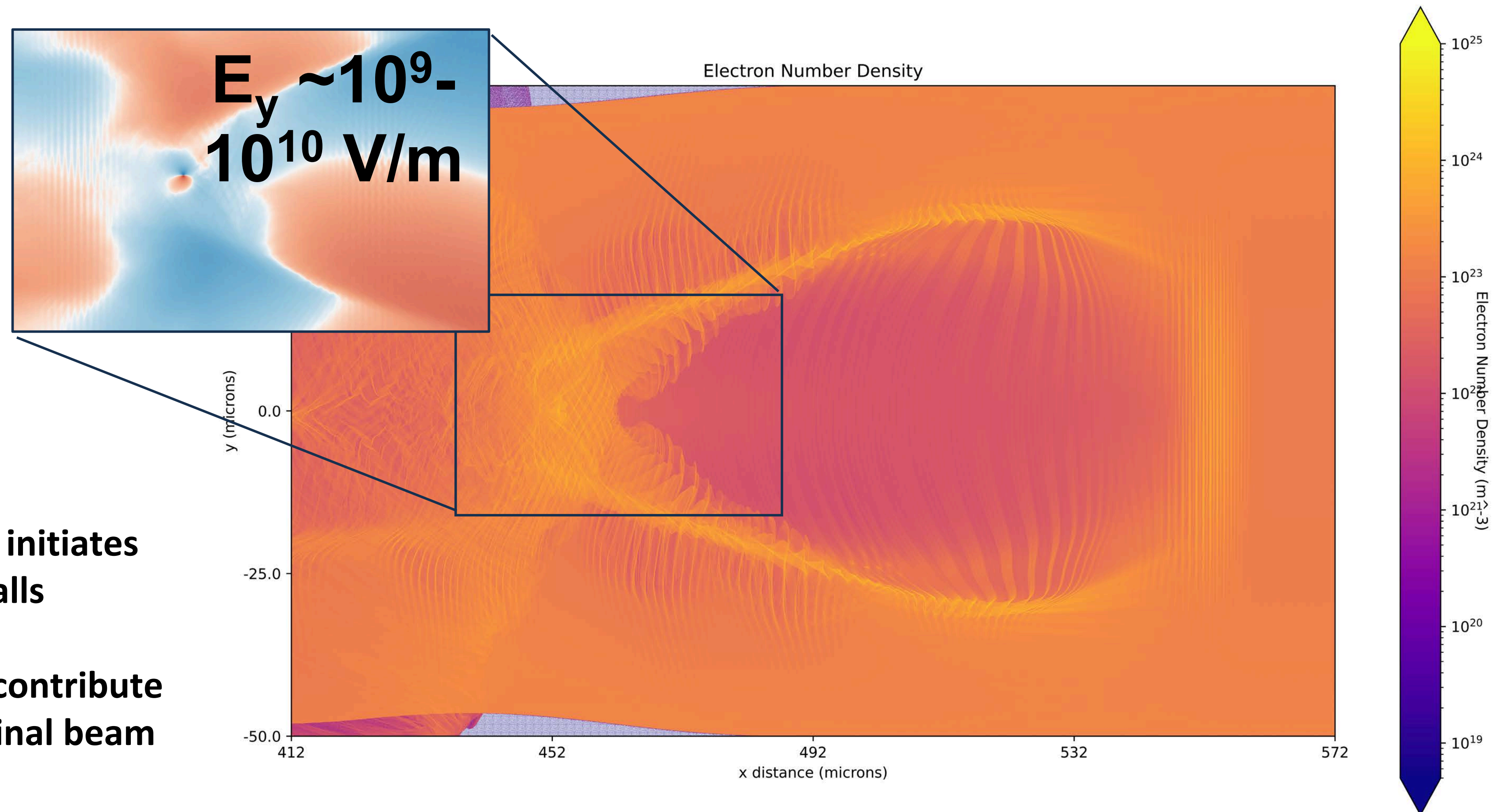
Hi-res simulations show nanoparticles trigger injection by perturbing bubble

Nanoparticle diameter~20-50nm requires very high resolution—2d simulations for now

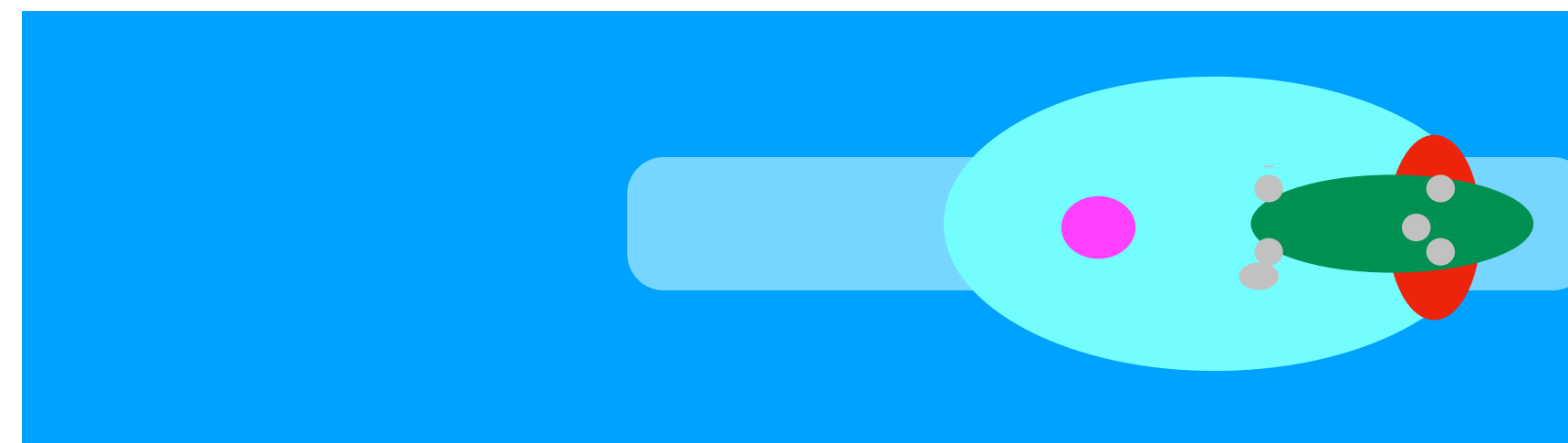
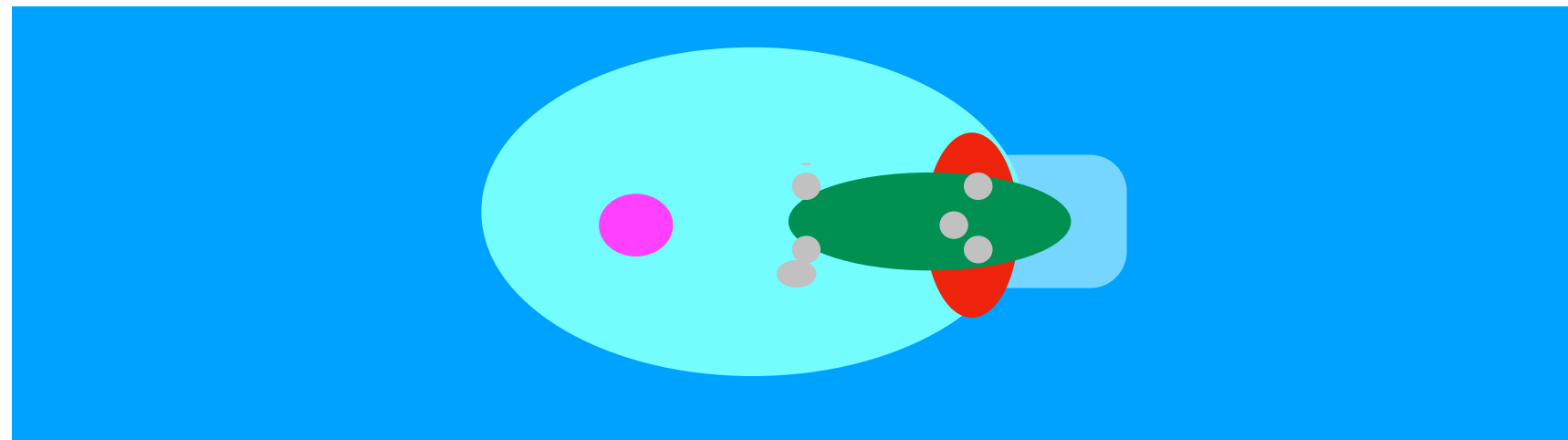
Nanoparticle fields similar to transverse bubble fields

Passage through bubble initiates perturbation to walls

Nanoparticle does **not** contribute majority of electrons to final beam



Hypothesis: hybrid Laser Plasma Wakefield Accelerator with nano-particle assisted injection



Laser-pulse and electron-bunch plasma wakefield accelerator

Tianhong Wang¹, Vladimir Khudik,^{1,2} and Gennady Shvets¹

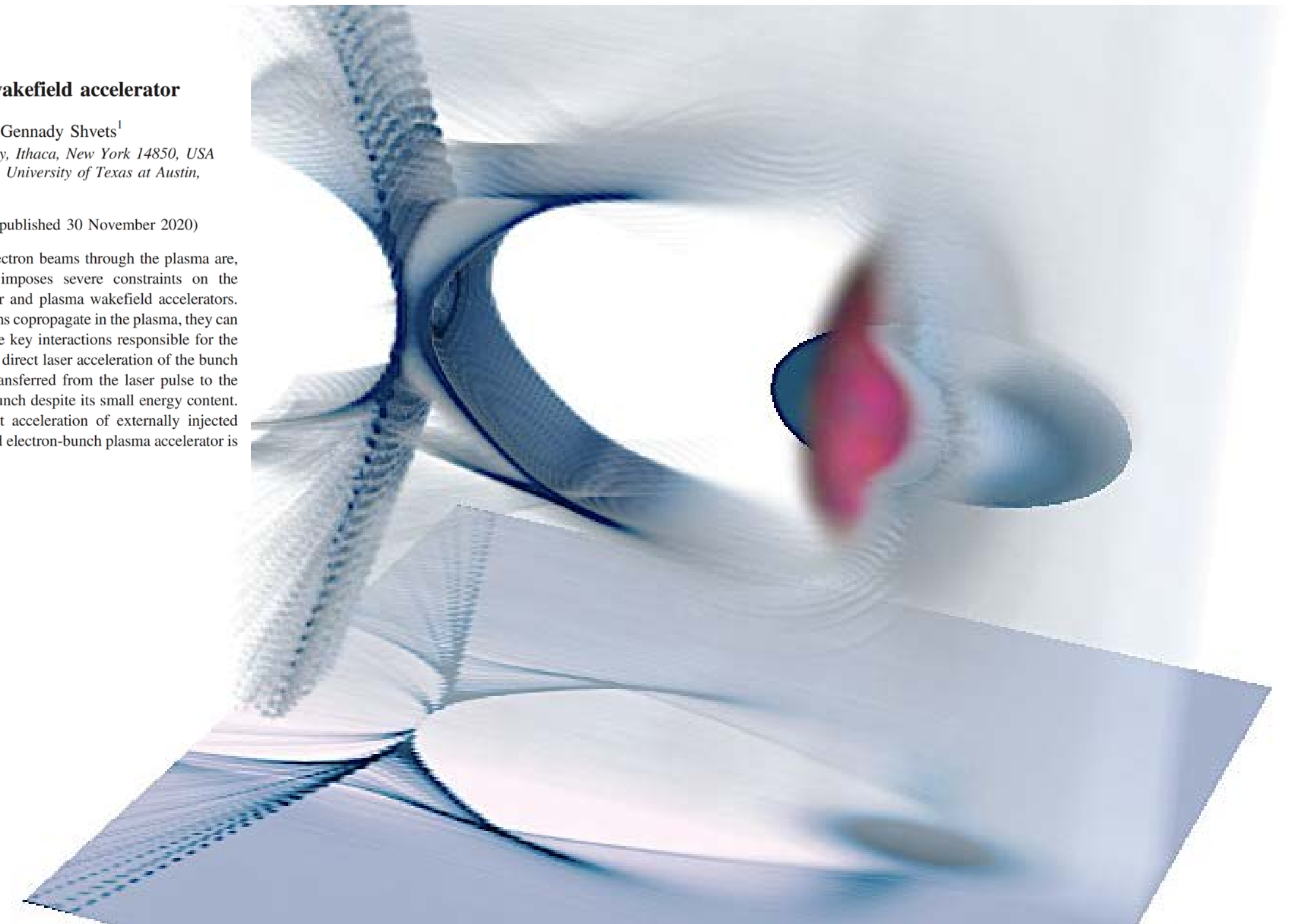
¹*School of Applied and Engineering Physics, Cornell University, Ithaca, New York 14850, USA*

²*Department of Physics and Institute for Fusion Studies, The University of Texas at Austin, Austin, Texas 78712, USA*

(Received 26 July 2020; accepted 11 November 2020; published 30 November 2020)

Propagation distances of intense laser pulses and high-charge electron beams through the plasma are, respectively, limited by diffraction and self-deceleration. This imposes severe constraints on the performance of the two major advanced accelerator concepts: laser and plasma wakefield accelerators. Using numerical simulations, we demonstrate that when the two beams copropagate in the plasma, they can interact synergistically and extend each other's travel distances. The key interactions responsible for the synergy are found to be laser channeling by the electron bunch, and direct laser acceleration of the bunch electrons by the laser pulse. Remarkably, the amount of energy transferred from the laser pulse to the plasma can be increased by several times by the guiding electron bunch despite its small energy content. Implications of such synergistic interactions for the high-gradient acceleration of externally injected witness charges are discussed, and a new concept of a laser-pulse and electron-bunch plasma accelerator is formulated.

DOI: 10.1103/PhysRevAccelBeams.23.111304



LEPA: Laser Pulse Electron Bunch Plasma Accelerator: T. Wang, V. Khudik, G. Shvets (2020)

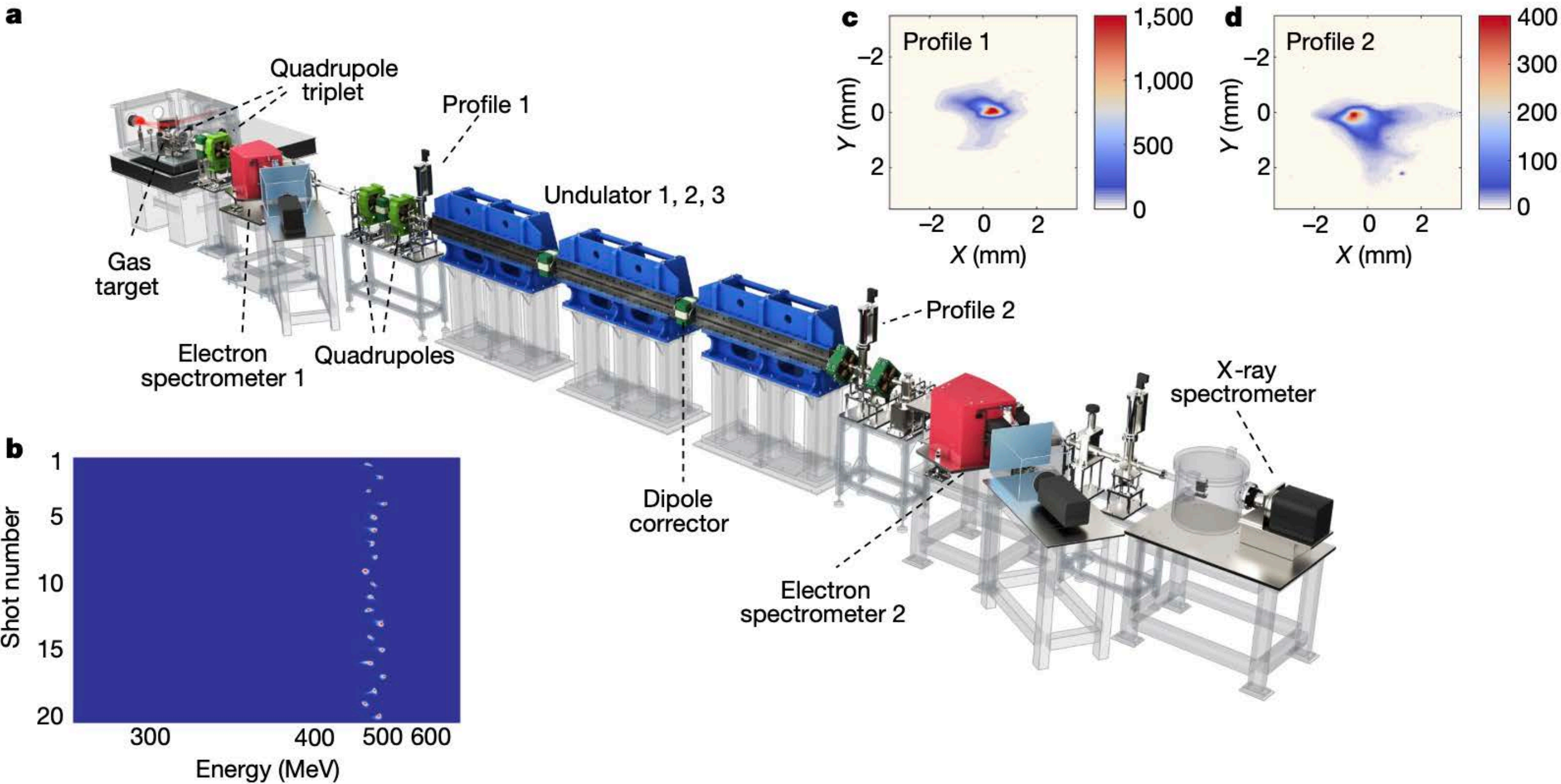
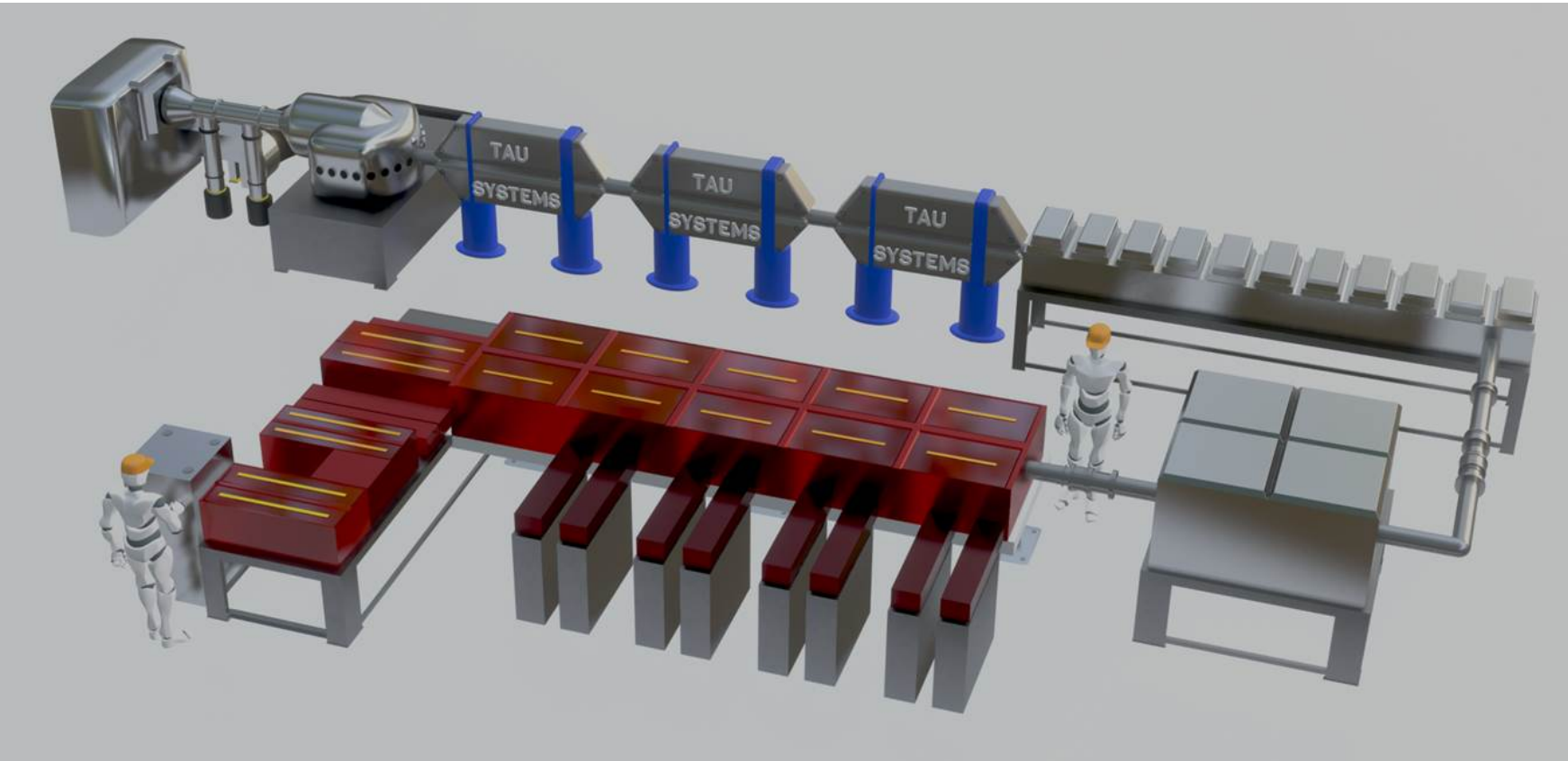
Free-electron lasing at 27 nanometres based on a laser wakefield accelerator

<https://doi.org/10.1038/s41586-021-03678-x>
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516 | Nature | Vol 595 | 22 July 2021

A recent article in Nature establishes the ability to achieve single pass FEL gain using beam from a reasonably stabilized LWFA beam.

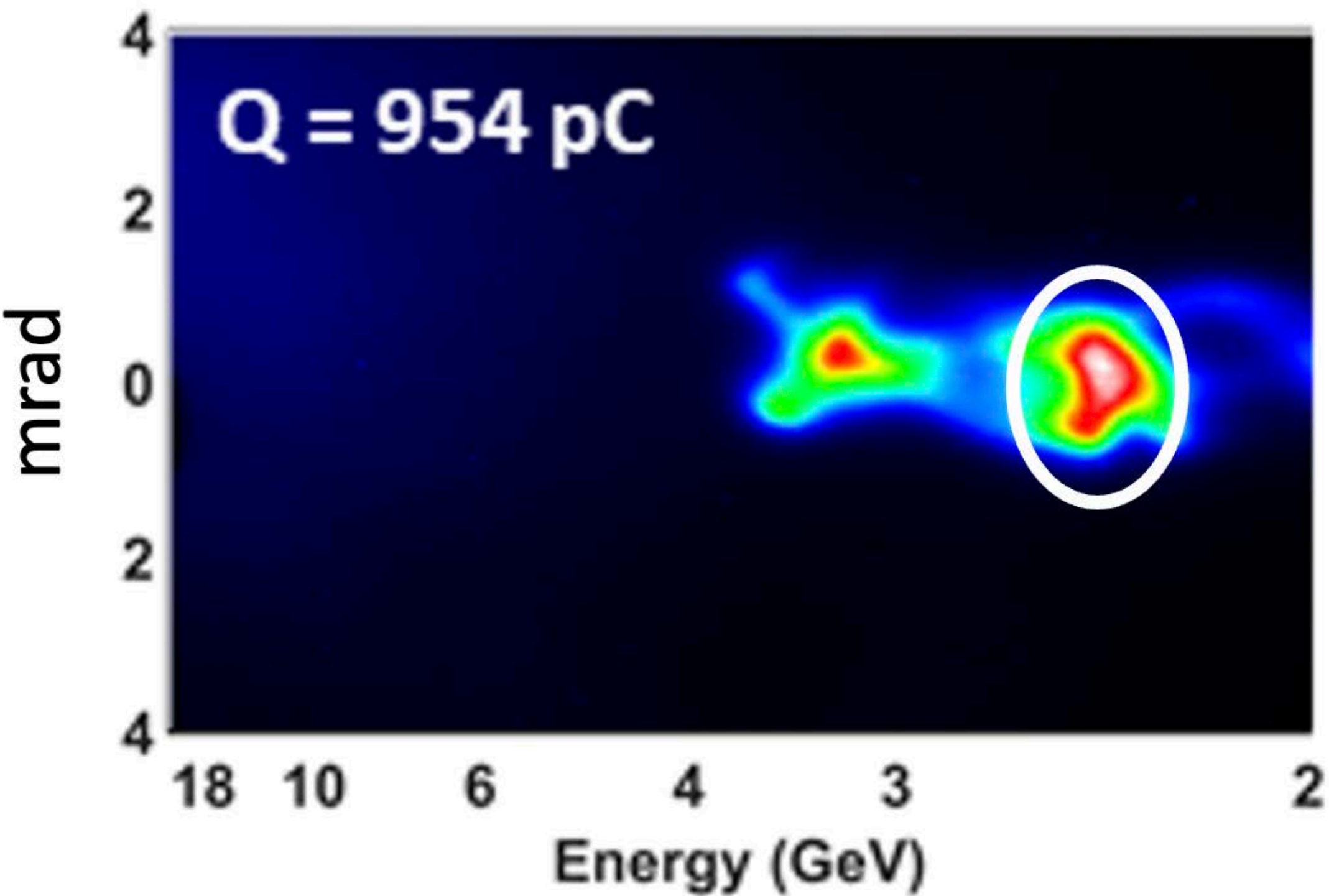


Texas Petawatt Beams: FEL Potential?

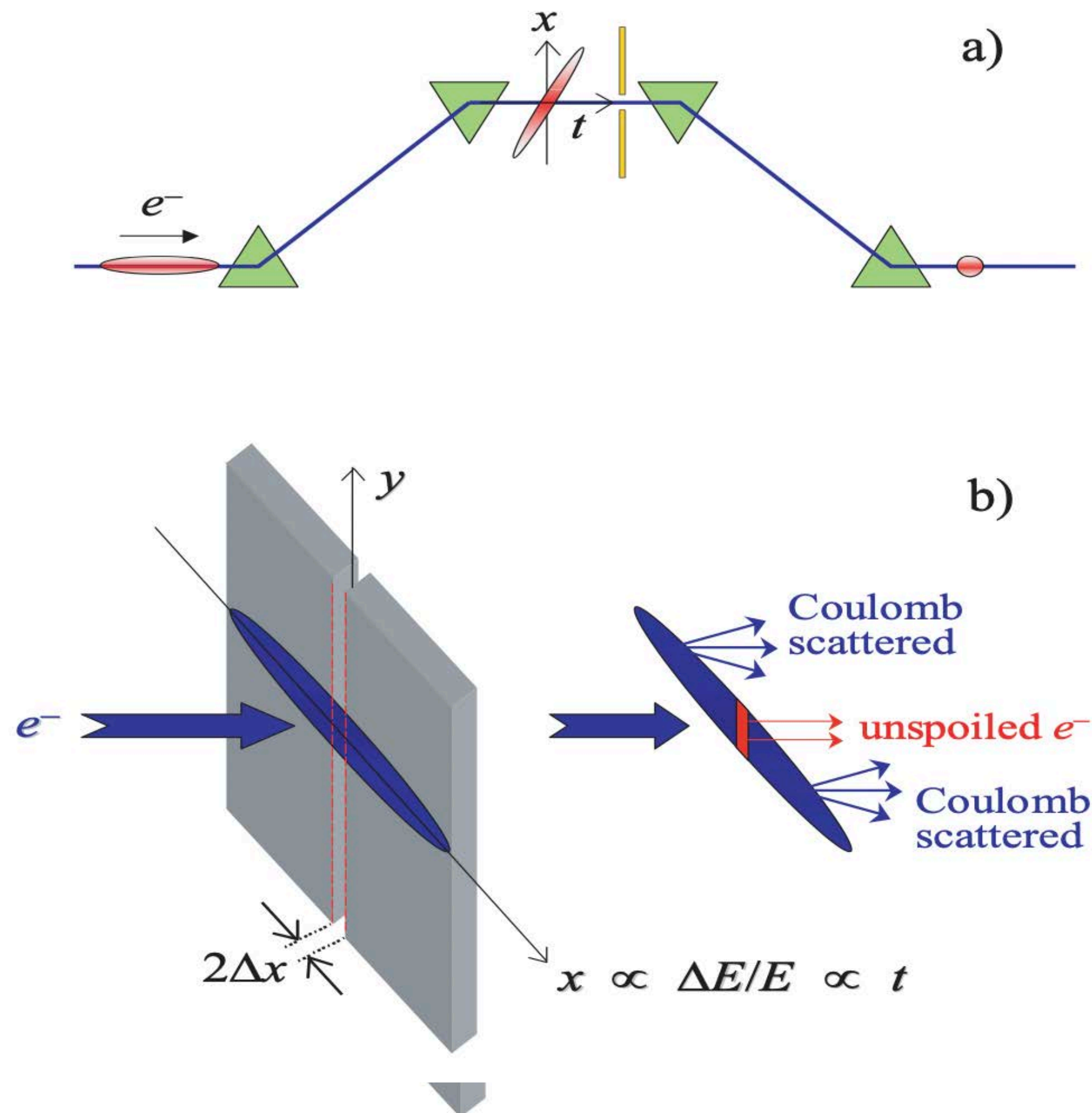
The Texas Petawatt Laser system can be used to generate beams with interesting possibilities, as the energies and charge delivered are significant.

The rough beam parameters for the shot shown in the white circle are given below. While certainly impressive, they are not quite suitable to drive a free-electron laser as the energy spread is too large. Additional beam manipulation must be done.

Quantity	Value
Energy [GeV]	2.3
rms Energy Spread [%]	2
Charge [pC]	800
rms Bunch Length [fs]	40
Geometric Emittance [nm-rad]	0.3



Beam Slicing to Improve the Beam Properties



If we add a dispersive section that has controllable R_{56} and can be made isochronous and dispersion free as necessary, and if the LWFA generated electron beam has small slice emittance, then one could slice out a small portion of the beam with reasonable energy spread. This portion of the bunch would have parameters suitable for single pass FEL operation.

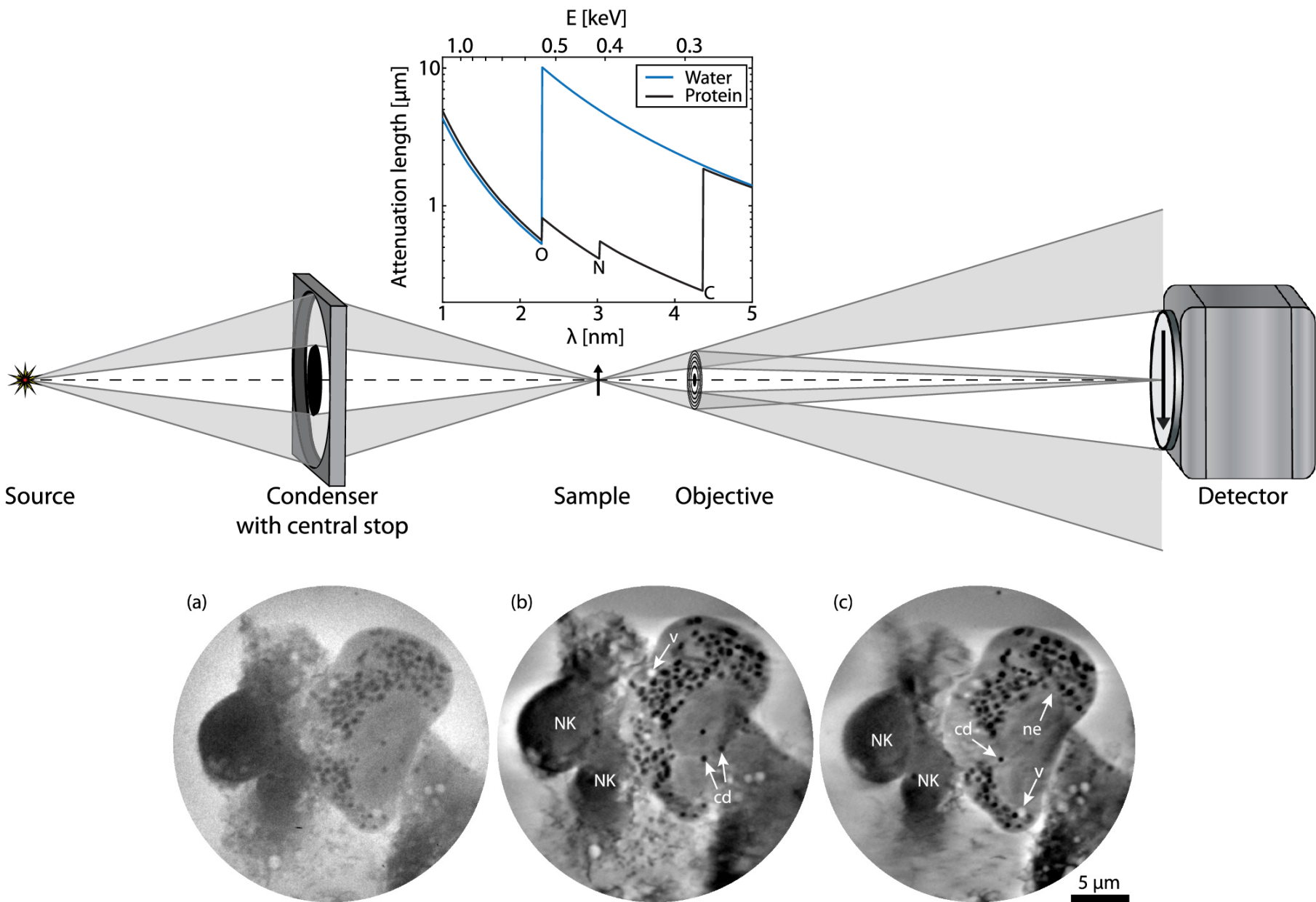
Femtosecond and sub-femtosecond x-ray pulses from a SASE-based free-electron laser

P. Emma, et al.

SLAC-PUB-10002



FEL Operation at the the Water Window



Slicing the beam in afore mentioned manner and using a parameterization for FEL performance* gives the following beam and FEL properties. Saturation in the water window is achieve with a 16.4-m long undulator

Quantity	Value
Energy [GeV]	2.3
rms Energy Spread [%]	0.5
Charge [pC]	200
rms Bunch Length [fs]	10
Geometric Emittance [nm-rad]	0.3
Undulator Period [cm]	2.78
Undulator K value	2.6
Resonant Wavelength [nm]	3.0
Resonant Photon Energy [eV]	413
Average Beta in Undulator [m]	1.5
FEL Rho Parameter	0.005
1d FEL Gain Length [m]	0.245
Eta: 3d beam quality reduction term	2.44
3d FEL Gain Length [m]	0.84
3d FEL Saturation Length [m]	16.4
3d Saturated Peak Power [GW]	32.4

Laboratory water-window x-ray microscopy

Mikael Kördel, Aurélie Dehlinger, Christian Seim, Ulrich Vogt, Emelie Fogelqvist, Jonas A. Sellberg, Holger Stiel, and Hans M. Hertz

Optica Vol. 7, Issue 6, pp. 658-674 (2020) • <https://doi.org/10.1364/OPTICA.393014>

Conclude: We may already have beam properties within regions of the TPW laser laser wakefield accelerated beams that could potentially drive a single pass FEL in the water window to saturation

DESIGN OPTIMIZATION FOR AN X-RAY
FREE ELECTRON LASER DRIVEN BY SLAC LINAC *

Ming Xie, Lawrence Berkeley Laboratory, Berkeley, CA 94720, USA



Nanoparticle-assisted electron injection into a wakefield at a 50 TW laser

www.nature.com/scientificreports

SCIENTIFIC REPORTS

nature research

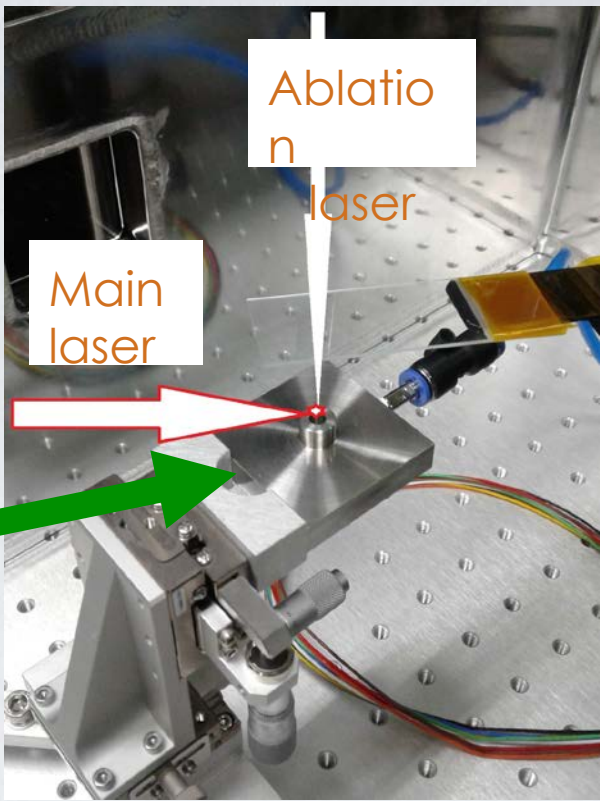
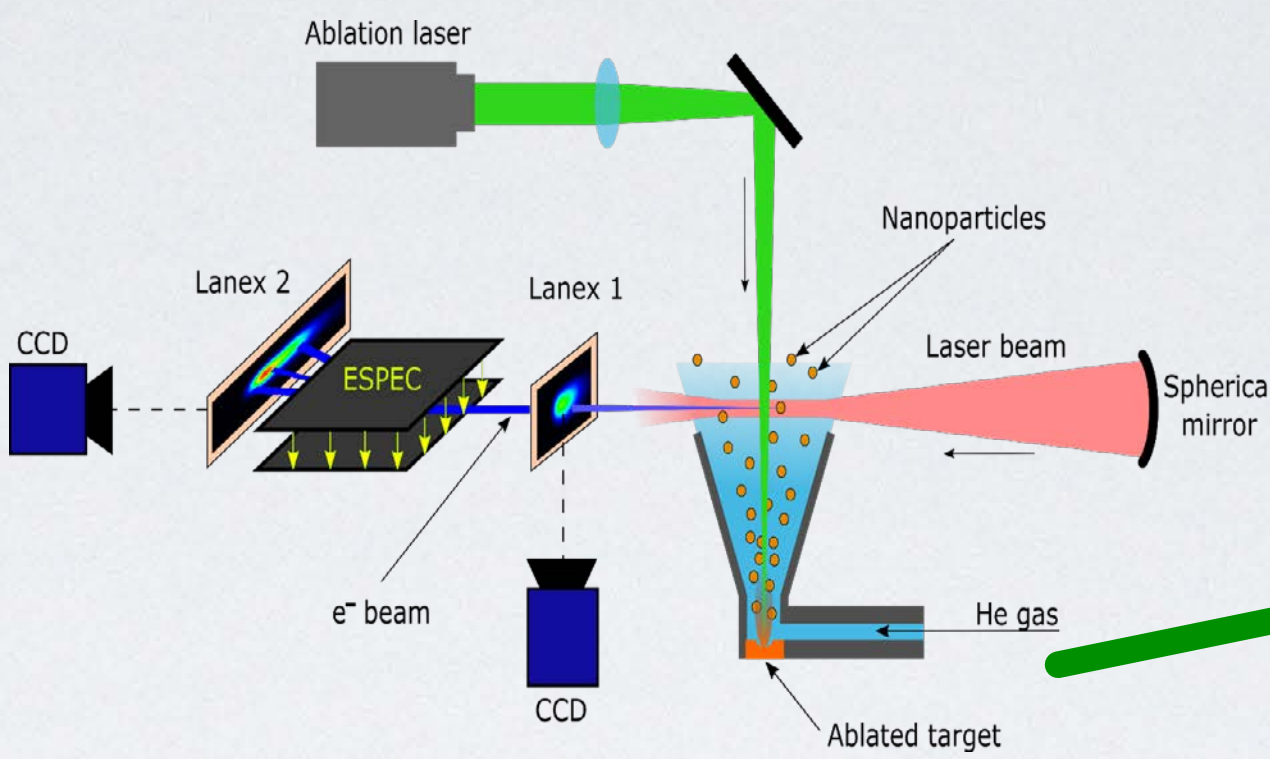
There are amendments to this paper

OPEN

Electron energy increase in a laser wakefield accelerator using up-ramp plasma density profiles

Received: 3 January 2019
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Constantin Aniculaesei¹, Vishwa Bandhu Pathak¹, Hyung Taek Kim^{1,2}, Kyung Hwan Oh¹, Byung Ju Yoo³, Enrico Brunetti⁴, Yong Ha Jang³, Calin Ioan Hojbota^{1,3}, Jung Hun Shin¹, Jong Ho Jeon¹, Seongha Cho¹, Myung Hoon Cho¹, Jae Hee Sung^{1,2}, Seong Ku Lee^{1,2}, Björn Manuel Hegelich^{1,3} & Chang Hee Nam^{1,3}



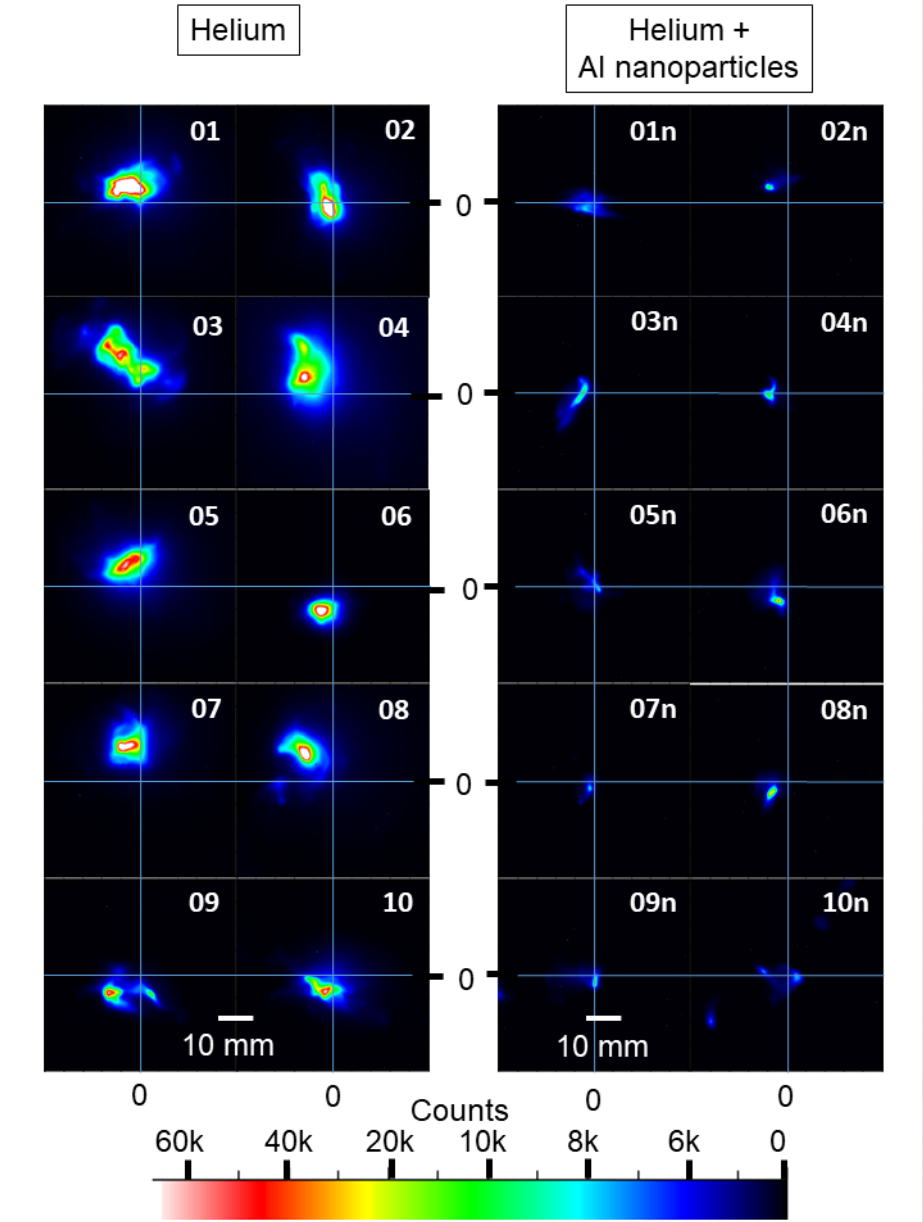
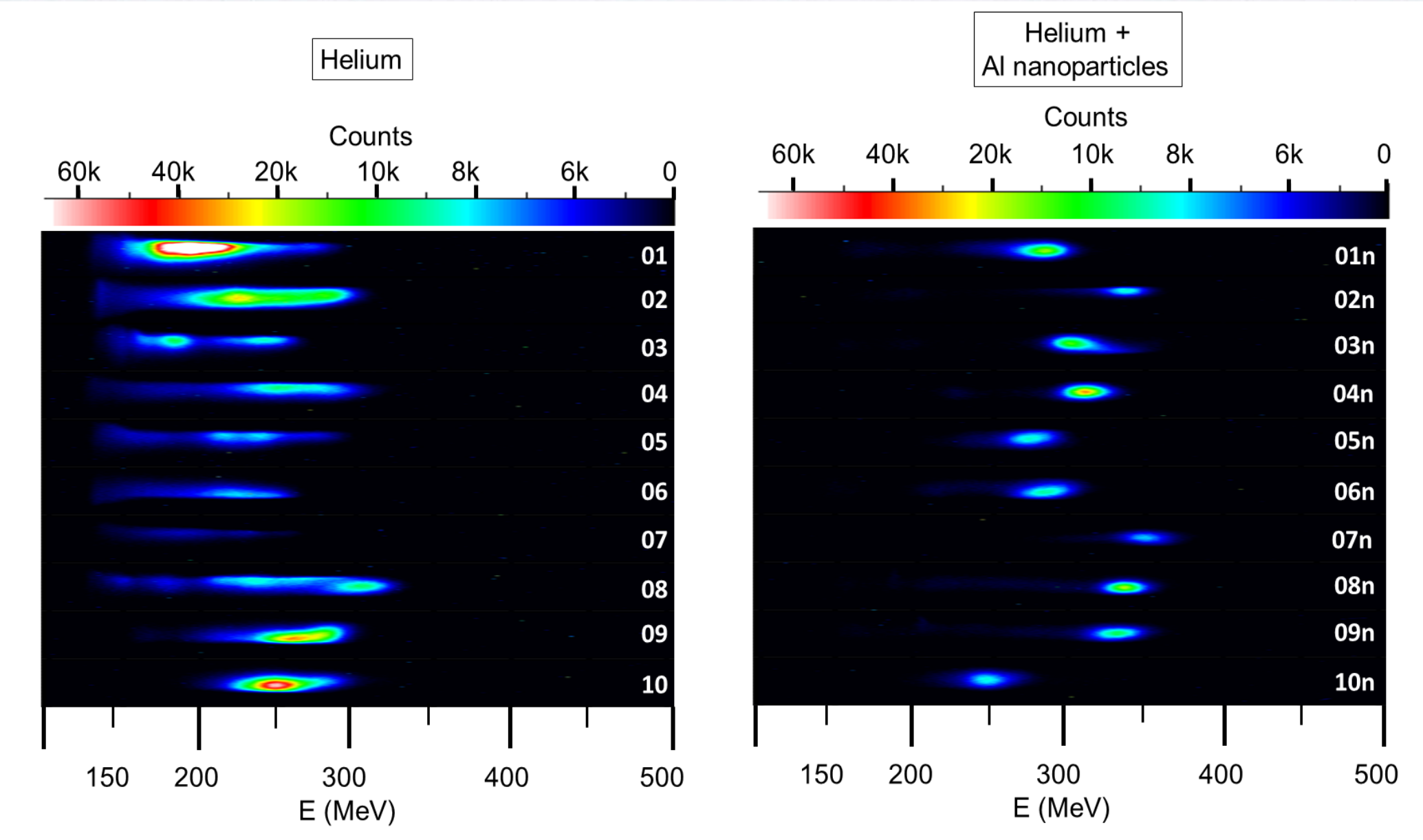
no
nanoparticles

with
nanoparticles

no
nanoparticles

with
nanoparticles

- The hybrid gas target can generate supersonic gas jets doped with any kind of nanoparticles
- The density and size of nanoparticle controlled by the laser energy, pulse width and fluence
- **Electron peak energy and energy spread greatly improved**
- **Electron beam divergence decreased**



Summary

- Wakefield Accelerators hold a huge potential to make accelerators and light sources ubiquitous and accessible
- Impressive recent progress
 - demonstration of PWFA with both electron and proton bunches (Facet, AWAKE)
 - staging of LWFA - PWFA (HZDR and others)
 - injection of RF accel. bunch into LWFA (Tsinghua)
 - >24h stable LWFA operation with ML (Des)
 - LWFA-driven FEL @ 27nm (SIOM)
 - >10 GeV, ~500s pC by nanoparticle assisted LPWFA (UT Austin)
- How to optimize emittance, energy spread, reproducibility simultaneously?
- What is the most promising laser-driver?